Graviton Scattering

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1 Introduction

The nature of gravity is one of the topics that raise the most questions in physics. Although we are able to describe three of the four fundamental interactions with a sophisticated formalism, gravity still resists such a description. General Relativity still agrees with the experiments it faces after more than one hundred years, while QFT has made predictions so precise that it cannot be ignored. To put this into perspective, QED predicts the g-factor of the electron so accurately that, if the factor itself were as big as New York City, the uncertainty would be as large as a hair [1]. For gravity, by contrast, a comparable test seems hopeless: the gravitational attraction between the electron and the proton would shift the phase of a hydrogen atom by only 43 arcseconds over a hundred times the age of the universe [2].

We decided to approach the subject of this thesis not by searching for more exotic and complicated physics capable of providing a full theory of gravitation, but rather by taking its most familiar part, the one we can most naturally quantize, and exploring how it interacts with nature, looking for signatures of quantum gravity. Modern quantum experiments allow for increasing control over density matrices and states. However, no experiment is capable of directly measuring the degrees of freedom of a graviton [**?**]. We aim to face the inverse problem: if no graviton can be directly measured, is there any way of understanding whether or not gravitons are real by looking at the particles we can measure? Starting from this, we address the problem of graviton-induced decoherence in a system [3]. To study it, we analyze the scattering of gravitons off matter in very simple systems, with the idea that every interaction couples matter degrees of freedom to gravitational ones and therefore leads to decoherence.

In Chapter **??**, we introduce linearized gravity, the most natural framework to quantize, explaining the different gauges and their meaning. In Chapter **??**, we carry out the quantization with various techniques, obtaining results that we could not find in the literature: we quantize the classically observable degrees of freedom, both directly and through covariant quantization. In Chapter **??**, we make use of the QFT formalism, deriving the basic ingredients of the scattering-matrix framework and showing differences and analogies with the photon case,

such as the appearance of a self-interaction term. In Chapter **??**, we start validating our results by computing the deflection of spin-n particles by a gravitational wave; in particular, we recover the classical geometrical results [4], plus some new terms that come from the particle-physics perspective, in which backward scattering is allowed. We then compute the analogue of Compton scattering for a very massive particle, which resembles the previous case, the difference being that we now also compute corrections to the velocity of the particle. We show that even a classical source of gravity can give rise to quantum behaviour. Finally, we compute scattering cross sections and their corrections. We also work out explicitly the angular part of the two-point function, a useful ingredient for computing decoherence in the Feynman–Vernon formalism [5], showing that it takes the form of spherical Bessel functions.

2 Introduction to Linearized Gravity

In this chapter, we will discuss how gravitational waves (GWs) emerge from general relativity and how they can be represented in a convenient way with a gauge choice, selecting the degrees of freedom they carry [6, 7]. We will also derive the Lagrangian densities of the theory, discussing how the interaction of linearized gravity with matter is modeled.

We will denote $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$ as the flat Minkowski metric.

2.1 Perturbation of the flat metric

In GR, we speak of *linearized gravity* whenever our metric is almost flat. When this happens, the dominant physics emerges from special relativity, and the bending of spacetime plays a secondary role. Mathematically, we show this by performing an expansion of the metric around flat space:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1. \tag{2.1}$$

The definition of $|h_{\mu\nu}| \ll 1$ depends on the norm we choose, but more importantly on the frame of reference we are in, leading to possible ambiguities. In fact, the perturbation $h_{\mu\nu}$ transforms from one observer to another: we generalize this by considering an infinitesimal family of coordinate transformations

$$x^\mu \to x'^\mu = x^\mu + \xi^\mu(x), \tag{2.2}$$

Under (2.2) the perturbation transforms as

$$\boxed{h_{\mu\nu} \to h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu}, \tag{2.3}$$

which defines a family of diffeomorphisms of linearized metrics if and only if the derivatives $|\partial_\mu \xi_\nu| \ll 1$ are at most of the same order of smallness as $|h_{\mu\nu}|$ (i.e. only if $|\partial_\mu \xi_\nu| \lesssim |h_{\mu\nu}|$ we remain in linearized gravity).

The linearized metric is a *tensor*. Under Lorentz transformations $h_{\mu\nu} \to \Lambda_\mu^{\ \rho} \Lambda_\nu^{\ \sigma} h_{\rho\sigma}(x)$, since

$$g_{\mu\nu}(x) \to g'_{\mu\nu}(x') = \Lambda_\mu^{\ \rho} \Lambda_\nu^{\ \sigma} g_{\rho\sigma}(x) = \Lambda_\mu^{\ \rho} \Lambda_\nu^{\ \sigma} \left[\eta_{\rho\sigma} + h_{\rho\sigma}(x)\right] = \eta_{\mu\nu} + \Lambda_\mu^{\ \rho} \Lambda_\nu^{\ \sigma} h_{\rho\sigma}(x). \tag{2.4}$$

We can think of linearized gravity as a *covariant* theory, as long as we start from a valid frame of reference and limit ourselves to slowly varying ones. Rotations never spoil (2.1), while boosts can. It will be convenient to rescale the perturbation as:

$$\boxed{h_{\mu\nu} \to \kappa h_{\mu\nu}} \tag{2.5}$$

2.2 Equations of Motion and Field Action

General Relativity is a field theory with action $S = S_E + S_M$, where:

$$S_E = \frac{2}{\kappa^2}\int d^4x\sqrt{-g}R, \qquad \kappa = \sqrt{32\pi G} \tag{2.6}$$

is the Einstein-Hilbert action and S_M is the matter action. Higher curvature proposals exist [8], but we will not consider them here. R, $R_{\mu\nu}$ and $R^{\rho\sigma}_{\mu\nu}$ are respectively the Ricci scalar, Ricci tensor and Riemann tensor. Varying the action with respect to the metric, we obtain *Einstein's Equations of Motion*:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G\, T_{\mu\nu} \tag{2.7}$$

We can have a convenient representation of the EoM by defining the trace and the trace-reversed perturbation:

$$h = \eta^{\mu\nu}h_{\mu\nu}, \qquad \bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h, \tag{2.8}$$

since we can find that the EoM can be linearized, we compute from (2.7) (further ref. Maggiore, GWs, chapter 1 [6]):

$$\boxed{\Box\bar{h}_{\mu\nu} + \eta_{\mu\nu}\partial_\rho\partial_\sigma\bar{h}^{\rho\sigma} - \partial_\rho\partial_\nu\bar{h}_\mu^{\ \rho} - \partial_\rho\partial_\mu\bar{h}_\nu^{\ \rho} = -\frac{\kappa}{2}T_{\mu\nu}.} \tag{2.9}$$

To prove this, we have performed a Taylor expansion of the geometrical components of (2.7). The metric is well defined, while its inverse can be found by imposing $g^{\mu\nu}g_{\nu\rho} = \delta^\mu_\rho$:

$$g^{\mu\nu} = \eta^{\mu\nu} - \kappa h^{\mu\nu} + \kappa^2 h^{\mu\lambda}h^\nu_\lambda - \kappa^3 h^{\mu\lambda}h^\alpha_\lambda h^\nu_\alpha + \dots, \tag{2.10}$$

We can also find some related quantities, such as the metric determinant, where we used $\ln\det g = \mathrm{tr}\ln g$:

$$\det(g_{\mu\nu}) = -1 - \kappa h + \tfrac{1}{2}\kappa^2(h^\mu_\rho h^\rho_\mu - h^2) + \tfrac{1}{6}\kappa^3(-2h^\mu_\rho h^\rho_\gamma h^\gamma_\mu + 3h\, h^\mu_\rho h^\rho_\mu - h^3) + \dots, \tag{2.11}$$

because from the logarithm expansion we had found that $\ln(\det(-g)) = \mathrm{tr}(\ln(-g))$ resulted in

$$\det g_{\mu\nu} = \det\eta_{\mu\alpha}\det\left(\delta^\alpha_\nu + \kappa h^\alpha_\nu\right), = (-1)\det(\mathbf{1} + \mathbf{H}). \tag{2.12}$$

From this equation one can find

$$\sqrt{-g} = 1 + \frac{\kappa}{2}h + \frac{\kappa^2}{8}(h^2 - 2h^\mu_\rho h^\rho_\mu) + \frac{\kappa^3}{48}(h^3 - 6h\, h^\mu_\rho h^\rho_\mu + 8h^\mu_\rho h^\rho_\gamma h^\gamma_\mu) + \dots. \tag{2.13}$$

To prove (2.9) we must compute more geometrical quantities expanded in powers of κ. With

some calculations we find:

$$\Gamma^{\lambda}_{\mu\nu} = \frac{\kappa}{2}\left(\eta^{\lambda\sigma} - \kappa h^{\lambda\sigma} + \kappa^2 h^{\lambda\alpha} h^{\sigma}_{\alpha}\right)\left(\partial_\mu h_{\sigma\nu} + \partial_\nu h_{\sigma\mu} - \partial_\sigma h_{\mu\nu}\right). \tag{2.14}$$

$$R = g^{\mu\nu}\left[\partial_\nu \Gamma^{\lambda}_{\mu\lambda} - \partial_\lambda \Gamma^{\lambda}_{\mu\nu} + \Gamma^{\tau}_{\mu\lambda}\Gamma^{\lambda}_{\tau\nu} - \Gamma^{\tau}_{\mu\nu}\Gamma^{\lambda}_{\tau\lambda}\right], \tag{2.15}$$

$$R^{(1)}_{\mu\nu} = \frac{\kappa}{2}\left(-\Box h_{\mu\nu} - \partial_\mu\partial_\nu h + \partial^\lambda\partial_\mu h_{\nu\lambda} + \partial^\lambda\partial_\nu h_{\mu\lambda}\right) \text{ and} \tag{2.16}$$

$$G^{(1)}_{\mu\nu} = -\frac{\kappa}{2}\left(\Box\bar{h}_{\mu\nu} + \eta_{\mu\nu}\partial^\lambda\partial^\rho\bar{h}_{\lambda\rho} - \partial^\lambda\partial_\mu\bar{h}_{\nu\lambda} - \partial^\lambda\partial_\nu\bar{h}_{\mu\lambda}\right). \tag{2.17}$$

These can be put into (2.7) to prove that its leading order is (2.9).

On the other hand, we are interested in making explicit the series of lagrangian densities resulting from $\mathcal{L}_E = \frac{2}{\kappa^2}\sqrt{-g}\,R$ [2]. We apply the same logic as before and we obtain at leading order the *massless Pauli-Fierz* lagrangian (2.18) and at the next order a lagrangian with three $h_{\mu\nu}$ (Ref. Choi, 1994 [9]):

$$\boxed{\mathcal{L}^0_E = -\partial_\rho h_{\mu\nu}\partial^\rho h^{\mu\nu} + 2\partial_\rho h_{\mu\nu}\partial^\nu h^{\mu\rho} - 2\partial_\nu h^{\mu\nu}\partial_\mu h + \partial_\mu h\partial^\mu h} \tag{2.18}$$

$$\boxed{\begin{aligned}\mathcal{L}^1_E = \kappa\Big[&\frac{1}{2}h^{\alpha}_{\beta}\partial^\mu h^{\beta}_{\alpha}\partial_\mu h - \frac{1}{2}h^{\alpha}_{\beta}\partial_\alpha h^{\mu}_{\nu}\partial^\beta h^{\nu}_{\mu} - h^{\alpha}_{\beta}\partial_\mu h^{\nu}_{\alpha}\partial^\mu h^{\beta}_{\nu} \\ &+ \frac{1}{4}h\,\partial^\beta h^{\mu}_{\nu}\partial_\beta h^{\nu}_{\mu} + h^{\beta}_{\mu}\partial_\nu h^{\alpha}_{\beta}\partial^\mu h^{\nu}_{\alpha} - \frac{1}{8}h\,\partial^\nu h\,\partial_\nu h\Big]\end{aligned}} \tag{2.19}$$

Actually, in our (2.19) we imposed $\partial_\mu h^{\mu\nu} = \frac{1}{2}\partial^\nu h$.

The matter lagrangian is expanded in the formalism of *minimal coupling*, where the flat metric is replaced by the curved metric, and partial derivatives are replaced by covariant derivatives. Other frameworks exist, but we won't consider them. As the matter field, we choose a massive, real scalar field ϕ, which obeys the Klein-Gordon EoM. Its leading-order lagrangian density is the free Klein-Gordon one:

$$\boxed{\mathcal{L}^0_M = \left(\frac{1}{2}\eta^{\mu\nu}\partial_\nu\phi\partial_\mu\phi - \frac{\phi^2 m^2}{2}\right),} \tag{2.20}$$

while its first-order perturbation relates $h_{\mu\nu}$ to the flat-space stress-energy tensor:

$$\boxed{\mathcal{L}^1_M = -\frac{\kappa}{2}\,h^{\mu\nu}T_{\mu\nu} = -\frac{\kappa}{2}\,h^{\mu\nu}\left[\partial_\mu\phi\,\partial_\nu\phi - \eta_{\mu\nu}\left(\frac{1}{2}\partial_\rho\phi\,\partial^\rho\phi - \frac{1}{2}m^2\phi^2\right)\right]} \tag{2.21}$$

This is a common fact in field theories, where fields interact when one is coupled to a conserved

current of the other. In flat space, the stress-energy tensor is the *natural rank-2 conserved current associated with translation invariance*. It can be derived from Noether's theorem, from the translation invariance of the action. It's worth stressing that, in curved space, it's not true that $\partial_\mu T^{\mu\nu} = 0$; this is true at most with covariant derivatives. We express one further term:

$$\boxed{\mathcal{L}_M^2 = \kappa^2 \left\{ \frac{1}{4}\left(\frac{1}{2}h^2 - h_{\alpha\beta}h^{\alpha\beta}\right)\mathcal{L}_M - \frac{hh^{\mu\nu}}{4}\partial_\mu\phi\partial_\nu\phi + \frac{1}{2}h^{\mu\alpha}h_\alpha^\nu\partial_\mu\phi\partial_\nu\phi \right\}.} \tag{2.22}$$

2.3 Gauge Freedom and the GWs

Recalling the brief discussion of the *covariance principle* in linearized gravity (Sec. 1.1), we now show how it gives rise to a gauges. A generic change of coordinates introduces parameters $\xi(x)$, which allow us to modify $h_{\mu\nu}$ according to (2.3). This mechanism resembles a gauge freedom. We consider two gauge choices. The first one is called ***De Donder*** gauge [6]:

$$\boxed{\partial^\mu \bar{h}_{\mu\nu} = 0.} \tag{2.23}$$

We can always choose this gauge. Under (2.3)

$$\partial^\mu \bar{h}_{\mu\nu} \to \partial^\mu \bar{h}_{\mu\nu} - \Box\xi_\nu, \tag{2.24}$$

and calling $\partial^\nu \bar{h}_{\mu\nu} \equiv f_\mu(x)$, we can choose a gauge parameter such that $\Box\xi_\mu = f_\mu(x)$, thereby obtaining $\partial^\nu \bar{h}'_{\mu\nu} = 0$. Since the D'Alembertian operator is invertible with a Green's function procedure, we can always solve this equation.

The strength of this gauge is that the EoM reduce to:

$$\Box\bar{h}_{\mu\nu} = -\frac{\kappa}{2}T_{\mu\nu} \tag{2.25}$$

which implies stress-energy conservation

$$\partial^\mu T_{\mu\nu} = 0. \tag{2.26}$$

Even more interesting is that, in the absence of sources, we obtain the wave equation:

$$\boxed{\Box\bar{h}_{\mu\nu} = 0} \tag{2.27}$$

The De Donder gauge puts four constraints on a symmetric rank-2 tensor, leaving six degrees of freedom.

The harmonic condition (2.23) does not fix the gauge completely. Under (2.3), the trace-reversed field transforms as

$$\bar{h}_{\mu\nu} \;\rightarrow\; \bar{h}_{\mu\nu} + \xi_{\mu\nu}, \qquad \xi_{\mu\nu} \equiv \partial_\mu \xi_\nu + \partial_\nu \xi_\mu - \eta_{\mu\nu}\, \partial_\rho \xi^\rho . \tag{2.28}$$

This is interesting because, taking the divergence, $\partial^\mu \xi_{\mu\nu} = \Box \xi_\nu$, hence the De Donder condition is preserved by every further transformation satisfying:

$$\Box \xi_\mu = 0. \tag{2.29}$$

All the further ξ_μ which obey (2.29) allow for further gauge choices. *Outside the source* we can fully exploit this freedom: if

$$\Box \bar{h}_{\mu\nu} = \Box \xi_{\mu\nu} = 0, \tag{2.30}$$

we have four more functions that can gauge $h_{\mu\nu}$. We call ***transverse-traceless (TT)*** the choice where the following conditions are satisfied:

$$\boxed{h^{0\mu} = 0, \qquad h^i{}_i = 0, \qquad \partial_i h^{ij} = 0.} \tag{2.31}$$

For the explicit computation, see (Ref. Maggiore, GWs, Chapt.1 [6]).

Here, since $\bar{h}_{\mu\nu} = h_{\mu\nu}$, the equations of motion are

$$\boxed{\Box h_{\mu\nu} = 0.} \tag{2.32}$$

As discussed above, they are valid outside the sources only.

We can compute the form of such a perturbation by setting the wave momentum along z: from (2.31), $\epsilon_{3j} = 0$ and $\epsilon_{22} = -\epsilon_{11}$.

$$\boxed{h_{ij}^{\mathrm{TT}}(t,z) = \begin{pmatrix} h_+ & h_\times \\ h_\times & -h_+ \end{pmatrix}_{(x,y)} \cos\big[\omega(t-z)\big],} \tag{2.33}$$

with h_+ and $h_\times$ the amplitudes of the two physical polarizations.

From another perspective, a perturbation $h_{\mu\nu}$ propagating outside the sources, expressed in the De Donder gauge, can be projected onto the stricter TT gauge by introducing the transverse projector (Ref. Maggiore, GWs, Chapt. 1 [6]):

$$P_{ij}(\hat{n}) = \delta_{ij} - n_i n_j, \tag{2.34}$$

and using it to build the *Lambda tensor*

$$\Lambda_{ij,kl}(\hat{n}) = \frac{1}{2}\big(P_{ik}P_{jl} + P_{il}P_{jk} - P_{ij}P_{kl}\big), \tag{2.35}$$

which is a projector, $\Lambda_{ij,kl}\Lambda_{kl,mn} = \Lambda_{ij,mn}$, is transverse on every index, and traceless in both pairs, $\Lambda_{ii,kl} = \Lambda_{ij,kk} = 0$. The TT wave is then simply

$$\boxed{h_{ij}^{\rm TT} = \Lambda_{ij}^{kl}\, h_{kl}}, \tag{2.36}$$

Actually, the same operation defines the TT part $S_{ij}^{\rm TT} = \Lambda_{ij,kl}S_{kl}$ of any symmetric tensor.

We stress that the TT gauge can only be imposed in vacuum. Inside a source, where $\Box\bar{h}_{\mu\nu} \propto T_{\mu\nu} \neq 0$, the residual transformations (2.29) are no longer sufficient to remove the components $h_{0\mu}$ and h.
Due to this, the TT gauge captures the radiative degrees of freedom but not the Newtonian part of the interaction.

Summary: Lagrangians

For further use, we collect the results above:

$$\mathcal{L}_E^0 = -\partial_\rho h_{\mu\nu}\partial^\rho h^{\mu\nu} + 2\partial_\rho h_{\mu\nu}\partial^\nu h^{\mu\rho} - 2\partial_\nu h^{\mu\nu}\partial_\mu h + \partial_\mu h\partial^\mu h \tag{2.37}$$

$$\mathcal{L}_M^0 = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2, \tag{2.38}$$

$$\begin{aligned}\mathcal{L}_E^1 = \kappa\Big[&\frac{1}{2}h_\beta^\alpha\partial^\mu h_\alpha^\beta\partial_\mu h - \frac{1}{2}h_\beta^\alpha\partial_\alpha h_\nu^\mu\partial^\beta h_\mu^\nu - h_\beta^\alpha\partial_\mu h_\alpha^\nu\partial^\mu h_\nu^\beta \\ &+ \frac{1}{4}h\,\partial^\beta h_\nu^\mu\partial_\beta h_\mu^\nu + h_\mu^\beta\partial_\nu h_\beta^\alpha\partial^\mu h_\alpha^\nu - \frac{1}{8}h\,\partial^\nu h\,\partial_\nu h\Big],\end{aligned} \tag{2.39}$$

$$\mathcal{L}_M^1 = -\frac{\kappa}{2}h^{\mu\nu}\Big[\partial_\mu\phi\,\partial_\nu\phi - \eta_{\mu\nu}\Big(\frac{1}{2}\partial_\rho\phi\,\partial^\rho\phi - \frac{1}{2}m^2\phi^2\Big)\Big], \tag{2.40}$$

$$\mathcal{L}_M^2 = \kappa^2\Big[\frac{1}{4}\Big(\frac{1}{2}h^2 - h_{\alpha\beta}h^{\alpha\beta}\Big)\mathcal{L}_M^0 - \frac{1}{4}h\,h^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi + \frac{1}{2}h^{\mu\alpha}h_\alpha^\nu\,\partial_\mu\phi\,\partial_\nu\phi\Big]. \tag{2.41}$$

3 Quantization of the Linearized Gravity Field

3.1 Introduction

In the canonical quantization of the spin-1 field, we could find that the A^0 component of the potential has no conjugate momentum, since the Lagrangian doesn't depend on $\partial_0 A^0$. For a massive (Proca) field this didn't turn out to be a problem: from the equations of motion we could fix A^0 in terms of the other fields,

$$A_0 = -\frac{1}{m^2}\vec{\nabla}\cdot\vec{E}, \tag{3.1}$$

this was enough to quantize the spatial components and all physical quantities turned out to be covariant (Ref. Greiner, *Field Quantization*, Chap. 7 [10]). However, for the photon field, since $m \to 0$, we coulnd't use this costraint: the vector field A^μ has two spurious degrees of freedom. They reflected gauge invariance of the theory, we found how different A^μ were physically equivalent if connected by a gauge transformation. In the canonical quantization of QED there are two common procedures to deal with this:

1. **Non-covariant gauge fixing**: one imposes Coulomb gauge, to eliminate the spurious degrees of freedom and quantizes the rest. This procedure, however, makes us lose manifest Lorentz invariance and is not convenient for systematically investigating the perturbative series.

2. **Covariant quantization**: all components of A^μ are treated in the same way. The Lagrangian of the field is modified so as to also give A^0 a conjugate momentum and, to eliminate the spurious degrees of freedom, conditions are imposed on the allowed states of the state space (Gubta-Bleuler method [10]).

Linearized gravity faces the same problem: $h_{0\mu}$ have no conjugate momentum [11]. In the following we quantize the graviton by analogy with QED in both ways: covariantly, in the de Donder gauge (Secs. 2.2–2.4), and non-covariantly, in the TT gauge (Sec. 2.5).

3.2 The Linearized Gravity Field in the de Donder Gauge

For the leading-order free Einstein–Hilbert Lagrangian, the conjugate momenta of the components $h_{0\mu}$ vanish (see (2.37)). In QED, Fermi solved the analogous problem by adding a *gauge-fixing term* to the Lagrangian, which gives A^0 a conjugate momentum. The same can be done for the graviton field:

$$\mathcal{L}_E'^{(0)} = \mathcal{L}_E^{(0)} - \frac{1}{2}\left(\partial_\mu \bar{h}^{\mu\nu}\right)\left(\partial^\rho \bar{h}_{\rho\nu}\right) \tag{3.2}$$

For such choice, the Lagrangian can be brought into a particularly simple form upon integration by parts in the action integral:

$$\mathcal{L}_E^{\prime(0)} = \frac{1}{2}\Big[\partial_\mu h_{\alpha\beta}\partial^\mu h^{\alpha\beta} - \frac{1}{2}\partial_\mu h\,\partial^\mu h\Big] \tag{3.3}$$

And the canonically conjugate momenta $\pi^{\mu\nu}$ of $h_{\mu\nu}$ are:

$$\pi^{\mu\nu} = \frac{\partial \mathcal{L}_E^{\prime(0)}}{\partial \dot{h}_{\mu\nu}} = \dot{h}^{\mu\nu} - \tfrac{1}{2}\eta^{\mu\nu}\dot{h} \tag{3.4}$$

Now the conjugate momenta resulting from $\mathcal{L}_E^{\prime(0)}$ are non-zero for every component of $h_{\mu\nu}$; nevertheless, $\mathcal{L}_E^{\prime(0)}$ coincides with $\mathcal{L}_E^{(0)}$ when the harmonic gauge condition $\partial_\mu \bar{h}^{\mu\nu} = 0$ is imposed.

The reason this choice is called de Donder gauge fixing is that (3.2) corresponds to the addition of a term which vanishes in the de Donder gauge and which yields the EoM $\Box \bar{h}_{\mu\nu} = 0$ even when $\partial_\mu \bar{h}^{\mu\nu} \neq 0$. In general, there is a whole class of gauge-fixing terms: any term of the form $-\frac{\zeta}{2}\,(\partial_\mu \bar{h}^{\mu\nu})\,(\partial^\rho \bar{h}_{\rho\nu})$ is admissible. In our case $\zeta = 1$, as in the Feynman gauge of QED; other values of ζ would have given different EoM.

Classically, the de Donder gauge can always be reached by a fixed gauge transformation, but in canonical quantization one cannot impose it as an operator identity: this would be inconsistent with the canonical commutation relations. We will later show how to deal with this problem.

Performing the Legendre transform $\mathcal{H} = \pi^{\mu\nu}\dot{h}_{\mu\nu} - \mathcal{L}$, we obtain the leading-order *Hamiltonian density of the free graviton field*:

$$\mathcal{H}_0^{\mathrm{PF}} = \frac{1}{2}\Big(\pi_{\mu\nu}\pi^{\mu\nu} - \tfrac{1}{2}\pi^2 + (\vec{\nabla} h_{\mu\nu})\cdot(\vec{\nabla} h^{\mu\nu}) - \tfrac{1}{2}(\vec{\nabla} h)^2\Big), \tag{3.5}$$

which resembles the structure of both QED and the scalar case. We have neglected terms of order $O(\kappa)$, i.e. cubic in $h_{\mu\nu}$; they will be taken into account in Section 3. We also compute the inverse relation for $h_{\mu\nu}$:

$$\dot{h}_{\mu\nu} = \pi_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}\pi. \tag{3.6}$$

3.3 Canonical quantization in the De Donder gauge

Before quantizing the field, we compare our procedure with the original one by Gupta (Ref. Gupta, Phys. Lett. 1952 [12]) and with the path-integral approach [2, 13, 14]. We write (3.2) as

$$\mathcal{L}_g^{(0)} = \tfrac{1}{2}\,\partial_\mu h_{\alpha\beta}\,G^{\alpha\beta\gamma\delta}\,\partial^\mu h_{\gamma\delta} = \tfrac{1}{2}\Big[\,\partial_\mu h_{\alpha\beta}\,\partial^\mu h^{\alpha\beta} - \tfrac{1}{2}\,\partial_\mu h\,\partial^\mu h\,\Big], \tag{3.7}$$

$$\boxed{G^{\alpha\beta\gamma\delta} = \tfrac{1}{2}\big(\eta^{\alpha\gamma}\eta^{\beta\delta} + \eta^{\alpha\delta}\eta^{\beta\gamma} - \eta^{\alpha\beta}\eta^{\gamma\delta}\big)} \tag{3.8}$$

where G, the DeWitt supermetric, performs the trace reversal on symmetric tensors:

$$\bar{h}^{\mu\nu} \equiv G^{\mu\nu\alpha\beta} h_{\alpha\beta} = h^{\mu\nu} - \tfrac{1}{2}\eta^{\mu\nu} h. \tag{3.9}$$

And it's its own inverse:

$$G^{\mu\nu\rho\sigma} G_{\rho\sigma\alpha\beta} = \mathbf{1}_{\mu\nu}{}^{\alpha\beta} = \tfrac{1}{2}\left(\delta^\alpha_\mu \delta^\beta_\nu + \delta^\alpha_\nu \delta^\beta_\mu\right) \tag{3.10}$$

Therefore, the conjugate momentum carries the supermetric:

$$\hat{\pi}^{\mu\nu} = \frac{\partial \mathcal{L}_g^{(0)}}{\partial \dot{h}_{\mu\nu}} = G^{\mu\nu\alpha\beta}\, \dot{\hat{h}}_{\alpha\beta} = \dot{\bar{\hat{h}}}^{\mu\nu}. \tag{3.11}$$

While from analogy with the scalar case, we would have had $-\partial_0 \hat{h}_{\mu\nu}$.

Since the momentu $\pi_{\mu\nu}$, the equations of motion $\Box \bar{h}^{\mu\nu} = 0$ and the gauge condition $\partial_\mu \bar{h}^{\mu\nu} = 0$ are all expressed in terms of $\bar{h}_{\mu\nu}$, Gupta quantized the trace-reversed field, with the same procedure as the Gupta–Bleuler quantization of QED. In the path-integral formalism, instead, the propagator follows from inverting the kinetic operator $K^{\alpha\beta\gamma\delta} = -G^{\alpha\beta\gamma\delta}\Box$, which results in:

$$\langle 0|\, T\left\{h_{\mu\nu}(x)\, h_{\alpha\beta}(y)\right\} |0\rangle = i\, G_{\mu\nu\alpha\beta}\, D_F(x-y), \tag{3.12}$$

where D_F is the Feynman propagator of a massless scalar field.

We instead quantize canonically the metric perturbation $h_{\mu\nu}$, which is the physically observable field, and show that the same results are recovered. This is consistent, since in vacuum $\Box \bar{h}^{\mu\nu} = 0$ implies, taking the trace

$$\eta_{\mu\nu}\Box \bar{h}^{\mu\nu} = \eta_{\mu\nu}\Box(h^{\mu\nu} - \frac{1}{2}\eta^{\mu\nu} h) = -\Box h = 0. \tag{3.13}$$

and hence also $h_{\mu\nu}$ obeys wave equation $\Box h^{\mu\nu} = 0$.

To quantize the theory, we impose the canonical equal-time commutation relations in the space of symmetric tensors:

$$\left[\hat{h}_{\mu\nu}(\vec{x},t),\, \hat{\pi}^{\alpha\beta}(\vec{y},t)\right] = i\, \mathbf{1}_{\mu\nu}{}^{\alpha\beta}\, \delta^{(3)}(\vec{x}-\vec{y}), \qquad \left[\hat{h}, \hat{h}\right] = \left[\hat{\pi}, \hat{\pi}\right] = 0. \tag{3.14}$$

Since G is its own inverse, the definition of the momentum can be inverted as $\dot{\hat{h}}_{\alpha\beta} = G_{\alpha\beta\rho\sigma}\, \hat{\pi}^{\rho\sigma}$, and (3.14) takes the equivalent form

$$\boxed{\begin{aligned} \left[\hat{h}_{\mu\nu}(\vec{x},t),\, \dot{\hat{h}}_{\alpha\beta}(\vec{y},t)\right] &= i\, G_{\mu\nu\alpha\beta}\, \delta^{(3)}(\vec{x}-\vec{y}), \\ \left[\hat{h}_{\mu\nu},\, \hat{h}_{\alpha\beta}\right] = \left[\dot{\hat{h}}_{\mu\nu},\, \dot{\hat{h}}_{\alpha\beta}\right] &= 0. \end{aligned}} \tag{3.15}$$

We remark why the gauge condition cannot be imposed as an operator identity, $\partial^\mu \hat{\bar{h}}_{\mu\nu} \neq 0$:

$$\left[\partial^\mu \hat{\bar{h}}_{\mu\nu}(\vec{x},t),\, \hat{h}_{\alpha\beta}(\vec{y},t)\right] = \left[\hat{\pi}_{0\nu}(\vec{x},t),\, \hat{h}_{\alpha\beta}(\vec{y},t)\right] = -\frac{i}{2}\,\delta^{(3)}(\vec{x}-\vec{y})\,\left(\eta_{0\alpha}\eta_{\nu\beta}+\eta_{0\beta}\eta_{\nu\alpha}\right) \neq 0. \tag{3.16}$$

In section 2.4 we will finally solve this problem.

3.3.1 Fourier Decomposition of the Field Operator

We now expand the operator $\hat{h}^{\mu\nu}$ into plane waves. To do so, we need a basis of symmetric polarization tensors, which we build from the spin-1 polarization vectors. These satisfy $\epsilon^\mu(\vec{k},a)\,\epsilon_\mu(\vec{k},c) = \eta_{ac}$, with $\eta_{ac} = \mathrm{diag}(1,-1,-1,-1)$, and span Minkowski space, $\mathrm{Span}_a(\epsilon_\mu(a)) = \mathbb{R}^{1,3}$. Since $\hat{h}^{\mu\nu}$ is symmetric, we take their symmetrized products:

$$\epsilon^{\mu\nu}(\vec{k},\lambda) = \frac{1}{\sqrt{2\,(1+\delta_{ab})}}\left(\epsilon^\mu(\vec{k},a)\,\epsilon^\nu(\vec{k},b)+\epsilon^\mu(\vec{k},b)\,\epsilon^\nu(\vec{k},a)\right), \tag{3.17}$$

where $\lambda = (a,b)$ is a polarization index. By construction, they are a *ten-dimensional orthonormal system*.

As in the spin-1 case, the basis respects *orthogonality realations* with respect to an indefinite metric:

$$\epsilon_{\mu\nu}(\vec{k},\lambda)\,\epsilon^{\mu\nu}(\vec{k},\lambda') = \mathrm{H}_{\lambda\lambda'}. \tag{3.18}$$

The metric is $\mathrm{H}_{\lambda\lambda} = +1$ for $\lambda = (0,0),(i,i),(i,j)$ and $\mathrm{H}_{\lambda\lambda} = -1$ for each $\lambda = (0,i)$. To prove this, we fix $\lambda = (a,b)$ and $\lambda' = (c,d)$, calling $N_\lambda = 1/\sqrt{2(1+\delta_{ab})}$ and hiding $\vec{k}$ dependence, we read

$$\begin{aligned}\epsilon_{\mu\nu}(\lambda)\,\epsilon^{\mu\nu}(\lambda') &= N_\lambda N_{\lambda'}\left(\epsilon_\mu(a)\epsilon_\nu(b)+\epsilon_\mu(b)\epsilon_\nu(a)\right)\left(\epsilon^\mu(c)\epsilon^\nu(d)+\epsilon^\mu(d)\epsilon^\nu(c)\right)\\ &= 2\,N_\lambda N_{\lambda'}\left(\eta_{ac}\,\eta_{bd}+\eta_{ad}\,\eta_{bc}\right),\end{aligned} \tag{3.19}$$

$$\epsilon_{\mu\nu}(\lambda)\,\epsilon^{\mu\nu}(\lambda') = \begin{cases} 0 & \lambda \neq \lambda', \\ \eta_{aa}\,\eta_{bb} & \lambda = \lambda',\ a \neq b, \\ 1 & \lambda = \lambda',\ a = b, \end{cases} \qquad \square$$

On the other hand, we write the *completeness relation* as

$$\sum_{\lambda=1}^{10} \mathrm{H}_{\lambda\lambda}\,\epsilon_{\mu\nu}(\vec{k},\lambda)\,\epsilon_{\alpha\beta}(\vec{k},\lambda) = \mathbf{1}_{\mu\nu\alpha\beta} \equiv \tfrac{1}{2}\left(\eta_{\mu\alpha}\eta_{\nu\beta}+\eta_{\mu\beta}\eta_{\nu\alpha}\right). \tag{3.20}$$

Had we used a basis of generic (non-symmetric) rank-2 tensors, we would have introduced extra degrees of freedom that the graviton does not have. For further reference on why we want a symmetric spin-2 tensor, see Maggiore, *Gravitational Waves*, Chap. 2 [6].

With this basis, the mode expansion of the graviton field reads

$$\boxed{\hat{h}_{\mu\nu}(x) = \int \frac{d^3k}{\sqrt{(2\pi)^3 2\omega_k}} \sum_{\lambda=1}^{10} \Big(\hat{a}_{\vec{k}\lambda}\, e^{-ik\cdot x} + \hat{a}^{\dagger}_{\vec{k}\lambda}\, e^{+ik\cdot x} \Big) \epsilon_{\mu\nu}(\vec{k},\lambda)} \tag{3.21}$$

and, from (3.4), the conjugate momentum is

$$\boxed{\hat{\pi}^{\mu\nu}(x) = -\,i \int \frac{d^3k\ \omega_k}{\sqrt{(2\pi)^3 2\omega_k}} \sum_{\lambda} \Big(\hat{a}_{\vec{k}\lambda}\, e^{-ik\cdot x} - \hat{a}^{\dagger}_{\vec{k}\lambda}\, e^{+ik\cdot x} \Big) G^{\mu\nu\alpha\beta} \epsilon_{\alpha\beta}(\vec{k},\lambda).} \tag{3.22}$$

To recover the $\hat{a}$ operator algebra we proceed by analogy with QED. A single mode of the $A^{\mu}(x)$ field is

$$A^{\mu}_{\vec{k},\lambda}(x) = \frac{1}{\sqrt{2\omega_k (2\pi)^3}}\, \epsilon^{\mu}(\vec{k},\lambda)\, e^{-ik\cdot x}; \tag{3.23}$$

we define the scalar product

$$\big(A, B\big) = i \int d^3x\ A^{\mu *}(x) \overset{\leftrightarrow}{\partial_0} B_{\mu}(x) \qquad \text{with} \qquad A \overset{\leftrightarrow}{\partial_0} B = A(\partial_0 B) - (\partial_0 A) B. \tag{3.24}$$

This scalar product isolates single annihilation operators of a field, in fact

$$\begin{aligned} \big(A_{\vec{k}'\lambda'}, A_{\vec{k}\lambda}\big) &= i \int d^3x\ \frac{1}{\sqrt{2\omega_{k'}(2\pi)^3}}\, \frac{1}{\sqrt{2\omega_k (2\pi)^3}} \\ &\quad \times\ \epsilon^{\mu}(\vec{k}',\lambda')\, e^{ik'\cdot x} \overset{\leftrightarrow}{\partial_0} \epsilon_{\mu}(\vec{k},\lambda)\, e^{-ik\cdot x} \\ &= \delta^3(\vec{k}' - \vec{k})\ \epsilon^{\mu}(\vec{k},\lambda')\, \epsilon_{\mu}(\vec{k},\lambda) = \delta^3(\vec{k}' - \vec{k})\, \eta_{\lambda\lambda'}, \end{aligned} \tag{3.25}$$

$$\big(A^{*}_{\vec{k}'\lambda'}, A^{*}_{\vec{k}\lambda}\big) = -\,\delta^3(\vec{k}' - \vec{k})\, \eta_{\lambda\lambda'} \tag{3.26}$$

and

$$\big(A_{\vec{k}'\lambda'}, A^{*}_{\vec{k}\lambda}\big) = \big(A^{*}_{\vec{k}'\lambda'}, A_{\vec{k}\lambda}\big) = 0. \tag{3.27}$$

prove that

$$\hat{a}_{\vec{k}\lambda} = \eta_{\lambda\lambda}\big(A_{\vec{k}\lambda}, \hat{A}\big) = i\, \eta_{\lambda\lambda} \int d^3x\ A^{\mu *}_{\vec{k}\lambda}(x) \overset{\leftrightarrow}{\partial_0} \hat{A}_{\mu}(x). \tag{3.28}$$

For the graviton field $\hat{h}^{\mu\nu}$, with modes $h^{\mu\nu}_{\vec{k}\lambda}(x) = \epsilon^{\mu\nu}(\vec{k},\lambda)\, e^{-ik\cdot x}/\sqrt{2\omega_k(2\pi)^3}$, the only difference in (3.25) is that we have the contraction $\epsilon_{\mu\nu}(\vec{k},\lambda)\, \epsilon^{\mu\nu}(\vec{k},\lambda') = \mathrm{H}_{\lambda\lambda'}$, so that $\hat{a}_{\vec{k}\lambda} = \mathrm{H}_{\lambda\lambda}\, (h_{\vec{k}\lambda}, \hat{h})$. Then

$$\boxed{a_{\vec{k}\lambda} = i\, \mathrm{H}_{\lambda\lambda} \int d^3x\ \frac{e^{\,ik\cdot x}}{\sqrt{2\omega_k (2\pi)^3}}\ \epsilon^{\mu\nu}(\vec{k},\lambda) \Big(\dot{h}_{\mu\nu}(x) - i\omega_k\, h_{\mu\nu}(x) \Big),} \tag{3.29}$$

or equivalently

$$\boxed{a_{\vec{k}\lambda} = i\,\mathrm{H}_{\lambda\lambda}\int d^3x\,\frac{e^{ik\cdot x}}{\sqrt{2\omega_k(2\pi)^3}}\left(\epsilon^{\mu\nu}(\vec{k},\lambda)\,G_{\mu\nu\alpha\beta}\,\pi^{\alpha\beta}(x) - i\omega_k\,\epsilon^{\mu\nu}(\vec{k},\lambda)\,h_{\mu\nu}(x)\right).} \tag{3.30}$$

Finally, using the canonical commutation relations, we obtain

$$\boxed{\left[\hat{a}_{\vec{k}\lambda},\hat{a}^\dagger_{\vec{k}'\lambda'}\right] = \mathcal{G}_{\lambda\lambda'}\,\delta^3(\vec{k}-\vec{k}'), \qquad \left[\hat{a}_{\vec{k}\lambda},\hat{a}_{\vec{k}'\lambda'}\right] = \left[\hat{a}^\dagger_{\vec{k}\lambda},\hat{a}^\dagger_{\vec{k}'\lambda'}\right] = 0,} \tag{3.31}$$

where we defined $\mathcal{G}_{\lambda\lambda'} = \mathrm{H}_{\lambda\lambda}\mathrm{H}_{\lambda'\lambda'}\,\epsilon^{\mu\nu}(\vec{k},\lambda)\,G_{\mu\nu\alpha\beta}\,\epsilon^{\alpha\beta}(\vec{k},\lambda')$. Its QED analogous was diagonal, graviton one isn't. If we fix $\lambda = (a,a)$, $\lambda' = (b,b)$; then the trace $\epsilon(\lambda) = \eta_{aa}$ and by definition $\mathrm{H}_{\lambda\lambda} = +1$. With some computation

$$\mathcal{G}_{(a,a)(b,b)} = \delta_{ab} - \tfrac{1}{2}\,\eta_{aa}\,\eta_{bb}. \tag{3.32}$$

We have just performed canonical quantization in the physically observable field, the price to do this is that commutation relations gain a non-trivial structure depending on their polarization index.

3.4 The Gupta–Bleuler Method

Finally, we address the problem of the gauge condition $\partial^\mu\hat{\bar{h}}_{\mu\nu} = 0$. We will see that this problem cannot be ignored, because it is strictly related to the fact that some states have negative norm:

$$\left\langle \vec{k},\lambda\,\middle|\,\vec{k}',\lambda'\right\rangle = \langle 0|\,\hat{a}_{\vec{k}\lambda}\hat{a}^\dagger_{\vec{k}'\lambda'}\,|0\rangle = \mathcal{G}_{\lambda\lambda'}\,\delta^3(\vec{k}-\vec{k}'), \tag{3.33}$$

hence the Fock space is not a Hilbert space.

Following the Gupta–Bleuler procedure [12, 10], we impose a *condition on the state vectors* $|\Phi\rangle$ of the Fock space. In particular we set,

$$\left(\partial^\mu\hat{h}_{\mu\nu} - \tfrac{1}{2}\partial_\nu\hat{h}\right)^{(+)}|\Phi\rangle = 0 \tag{3.34}$$

where we denoted with $(+)$ the positive-frequency, annihilation related part. This is a stronger condition than setting the *expectation value* of the de Donder condition to zero; it also guarantees that the expected number of negative-norm states vanishes.

In terms of annihilation operators, the condition reads

$$\int\frac{d^3k}{\sqrt{2\omega_k(2\pi)^3}}\,e^{-ik\cdot x}\sum_{\lambda=1}^{10}k^\mu\left(\epsilon_{\mu\nu}(\vec{k},\lambda) - \tfrac{1}{2}\,\eta_{\mu\nu}\,\epsilon^\alpha{}_\alpha(\vec{k},\lambda)\right)\hat{a}_{\vec{k}\lambda}\,|\Phi\rangle = 0, \tag{3.35}$$

and, since the plane waves are linearly independent, for each $\vec{k}$:

$$\sum_{\lambda=1}^{10}\Big(k^\mu \epsilon_{\mu\nu}(\vec{k},\lambda) - \tfrac{1}{2}\, k_\nu\, \epsilon^\alpha{}_\alpha(\vec{k},\lambda)\Big)\hat{a}_{\vec{k}\lambda}\,|\Phi\rangle = 0. \tag{3.36}$$

We can take $\vec{k} = \omega\hat{z}$.
For $\vec{k} = \omega\hat{z}$ the polarization vectors are:

$$\epsilon^\mu(\vec{k},0) = (1,0,0,0) \qquad \epsilon^\mu(\vec{k},1) = (0,1,0,0) \tag{3.37}$$

$$\epsilon^\mu(\vec{k},2) = (0,0,1,0) \qquad \epsilon^\mu(\vec{k},3) = (0,0,0,1) \tag{3.38}$$

so that $k_\nu = \omega\big(\epsilon_\nu(\vec{k},0) + \epsilon_\nu(\vec{k},3)\big)$.
Therefore, after some computations, we obtain four equations for the annihilation operators:

$$\big(\hat{a}_{11} + \hat{a}_{22}\big)\,|\Phi\rangle = 0, \tag{3.39a}$$

$$\big(\hat{a}_{01} - \hat{a}_{13}\big)\,|\Phi\rangle = 0, \tag{3.39b}$$

$$\big(\hat{a}_{02} - \hat{a}_{23}\big)\,|\Phi\rangle = 0, \tag{3.39c}$$

$$\big(\hat{a}_{00} - \sqrt{2}\,\hat{a}_{03} + \hat{a}_{33}\big)\,|\Phi\rangle = 0 \tag{3.39d}$$

The second and third equations have the same form as the QED Gupta–Bleuler condition $(\hat{a}_0 - \hat{a}_3)\,|\Phi\rangle = 0$.

Let us denote by $\hat{L}_{k\nu}$ the four combinations of annihilation operators appearing in (3.39), so that the physical states are defined by $\hat{L}_{k\nu}\,|\Phi\rangle = 0$. We find that:

- each $\hat{L}^\dagger_{k\nu}$ creates a zero-norm state (see the Appendix);
- the $\hat{L}_{k\nu}$ are mutually orthogonal, i.e. $[\hat{L}_{k\mu}, \hat{L}^\dagger_{k\nu}] = 0$ for every μ, ν;
- acting with $\hat{L}^\dagger_{k\nu}$ on a physical state gives another physical state: these are excitations that don't break the de Donder condition on expectation values.

Asking to have most of these excitations with zero expectation value (i.e. $\hat{n}_{11} + \hat{n}_{22} = 0$), we are left with two physical polarizations, which we call

$$\hat{a}_+ = \tfrac{1}{\sqrt{2}}\big(\hat{a}_{11} - \hat{a}_{22}\big), \qquad \hat{a}_\times = \hat{a}_{12}, \tag{3.40}$$

with polarization tensors

$$\epsilon^{(+)}_{\mu\nu} \propto \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \epsilon^{(\times)}_{\mu\nu} \propto \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{3.41}$$

Therefore, a physical state contains TT modes, plus possible excitations created by $\hat{L}^\dagger_{k\nu}$. However, they do not contribute to norms and expectation values, exactly as $\langle n_0 - n_3 \rangle = 0$ in QED.

One can show that the physical states form *equivalence classes*. As in QED, they are generated by gauge freedom. Denoting by $|\Phi_T\rangle$ a state containing only TT quanta, every state of its class can be written as

$$|\Phi_c\rangle = \hat{R}_c\,|\Phi_T\rangle\,, \qquad \hat{R}_c = \exp\Big(\int d^3k\; c^\nu(\vec{k})\,\hat{L}^\dagger_{k\nu}\Big) = 1 + \int d^3k\; c^\nu(\vec{k})\,\hat{L}^\dagger_{k\nu} + \dots, \quad (3.42)$$

where the four coefficient functions $c^\nu(\vec{k})$ descrive the states.

3.5 De Donder propagator

Finally we derive the propagator, by some machinery. Let us call:

$$C_{\mu\nu\alpha\beta}(x-y) = \big[\hat{h}_{\mu\nu}(x),\, \hat{h}_{\alpha\beta}(y)\big], \quad (3.43)$$

which can be computed directly from the plane-wave expansion of the field operator,

$$\hat{h}_{\mu\nu}(x) = \int \frac{d^3k}{\sqrt{2\omega_k(2\pi)^3}} \sum_{\lambda=1}^{10} \epsilon_{\mu\nu}(\vec{k},\lambda)\Big(\hat{a}_{\vec{k}\lambda}\, e^{-ik\cdot x} + \hat{a}^\dagger_{\vec{k}\lambda}\, e^{ik\cdot x}\Big). \quad (3.44)$$

Using (3.31):

$$C_{\mu\nu\alpha\beta}(x-y) = \int \frac{d^3k}{2\omega_k(2\pi)^3}\Big[\sum_{\lambda,\lambda'} \epsilon_{\mu\nu}(\vec{k},\lambda)\,\mathcal{G}_{\lambda\lambda'}\,\epsilon_{\alpha\beta}(\vec{k},\lambda')\Big]\Big(e^{-ik\cdot(x-y)} - e^{ik\cdot(x-y)}\Big). \quad (3.45)$$

From the definition of $\mathcal{G}_{\lambda\lambda'}$ and the completeness relation (3.20) we recover

$$\sum_{\lambda,\lambda'} \epsilon_{\mu\nu}(\lambda)\,\mathrm{H}_{\lambda\lambda}\,\epsilon^{\rho\sigma}(\lambda)\,G_{\rho\sigma\kappa\tau}\,\epsilon^{\kappa\tau}(\lambda')\,\mathrm{H}_{\lambda'\lambda'}\,\epsilon_{\alpha\beta}(\lambda') = \mathbf{1}_{\mu\nu}{}^{\rho\sigma}\,G_{\rho\sigma\kappa\tau}\,\mathbf{1}^{\kappa\tau}{}_{\alpha\beta} = G_{\mu\nu\alpha\beta}. \quad (3.46)$$

This contribute is independent from $\vec{k}$, then the remaining part is Δ, the Pauli–Jordan function from the Klein-Gordon field:

$$C_{\mu\nu\alpha\beta}(x-y) = G_{\mu\nu\alpha\beta}\int \frac{d^3k}{2\omega_k(2\pi)^3}\Big(e^{-ik\cdot(x-y)} - e^{ik\cdot(x-y)}\Big) = i\,G_{\mu\nu\alpha\beta}\,\Delta(x-y), \quad (3.47)$$

The solution of this problem is unique, one could perform a Taylor expansion around $\tau = t - t'$ to convince himself. We have found:

$$\big[\hat{h}_{\mu\nu}(x),\, \hat{h}_{\alpha\beta}(y)\big] = i\,G_{\mu\nu\alpha\beta}\,\Delta(x-y), \quad (3.48)$$

Therefore, by analogy with the Klein-Gordon field, we write everything in the momentum space:

$$\boxed{\langle 0|\, T\{\hat{h}_{\mu\nu}(x)\, \hat{h}_{\alpha\beta}(y)\}\, |0\rangle = G_{\mu\nu\alpha\beta}\, \Delta_F(x-y) = \int \frac{d^4k}{(2\pi)^4}\, \frac{i\, G_{\mu\nu\alpha\beta}}{k^2 + i\varepsilon}\, e^{-ik\cdot(x-y)}.} \tag{3.49}$$

Further references for those computations in (Greiner, Field Quantization, 4.105 [10])

3.6 Canonical Quantization in the Transverse-Traceless Gauge

If we are prepared to lose manifest covariance, we can use the TT conditions (2.31). In this case the propagator takes a more complicated form, but the interaction vertices become much simpler. For a free field $h^{\mu\nu}$, the gauge can be chosen so that

$$h = 0, \qquad h^{0\mu} = 0, \qquad \partial_i h^{ij} = 0.$$

As in QED, this leads to a *radiation gauge*, meaning that we quantize only the propagating degrees of freedom of the theory. If we naively quantize the graviton field, however, we fail to recover the TT gauge:

$$[\partial_i h^{ij}(\vec{x}), \pi^{kl}(\vec{x}')] = \partial_i [h^{ij}(\vec{x}), \pi^{kl}(\vec{x}')] = i\, \partial_i \delta^{(3)}(\vec{x} - \vec{x}')\, \mathbf{1}^{ij,kl} \neq 0, \tag{3.50}$$

We therefore modify the ETCR so that they are compatible with the TT condition. In analogy with the Coulomb gauge in QED, we use the projector (2.35) to define the *transverse-traceless delta function*

$$\delta^{(3)}_{TT\, ij,kl}(\vec{x} - \vec{x}') = \int \frac{d^3k}{(2\pi)^3}\, \Lambda_{ij,kl}(\hat{k})\, e^{i\vec{k}\cdot(\vec{x}-\vec{x}')}. \tag{3.51}$$

By construction, this distribution satisfies all the TT conditions, and we impose

$$[h_{ij}(\vec{x}, t), \pi_{kl}(\vec{x}', t)] = i\, \delta^{(3)}_{TT\, ij,kl}(\vec{x} - \vec{x}'). \tag{3.52}$$

In this way the quantization is compatible with the TT condition. In momentum space the projector (2.35) is just a multiplicative factor, so the procedure is the same as before.

The plane-wave expansion of the field operator is particularly simple, since it contains only the two TT modes:

$$\hat{h}_{ij}(\vec{x}, t) = \int \frac{d^3k}{\sqrt{2\omega_k (2\pi)^3}} \sum_{\lambda=+,\times} \epsilon_{ij}(\vec{k}, \lambda) \big(\hat{a}_{\vec{k}\lambda} e^{-ik\cdot x} + \text{h.c.}\big). \tag{3.53}$$

As expected, the creation and annihilation operators satisfy a bosonic algebra. To avoid

repetition, we only report the final results:

$$\hat{a}_{\vec{k}\lambda} = i \int \frac{d^3x}{\sqrt{(2\pi)^3 2\omega_k}}\, e^{ik\cdot x}\, \epsilon^{ij}(\vec{k},\lambda)\Big(\hat{\pi}_{ij}(x) - i\omega_k \hat{h}_{ij}(x)\Big), \tag{3.54}$$

$$\hat{a}^\dagger_{\vec{k}\lambda} = -i \int \frac{d^3x}{\sqrt{(2\pi)^3 2\omega_k}}\, e^{-ik\cdot x}\, \epsilon^{ij}(\vec{k},\lambda)\Big(\hat{\pi}_{ij}(x) + i\omega_k \hat{h}_{ij}(x)\Big), \tag{3.55}$$

and the commutation relations are now diagonal:

$$\boxed{\big[\hat{a}_{\vec{k}\lambda},\, \hat{a}^\dagger_{\vec{k}'\lambda'}\big] = \delta_{\lambda\lambda'}\, \delta^{(3)}(\vec{k}-\vec{k}')} \tag{3.56}$$

Consequently, after normal ordering, the Hamiltonian is a sum of number operators:

$$H = \int d^3k\ \omega_k \sum_{\lambda=+,\times} \hat{n}_{\vec{k}\lambda}. \tag{3.57}$$

The Feynman propagator in the TT gauge follows from the sum over polarization tensors,

$$\sum_{\lambda=+,\times} \epsilon_{ij}(\vec{k},\lambda)\, \epsilon_{kl}(\vec{k},\lambda) = \frac{1}{2}\Big(P_{ik}P_{jl} + P_{il}P_{jk} - P_{ij}P_{kl}\Big) \equiv \Lambda_{ij,kl}(\hat{k}), \tag{3.58}$$

and reads

$$\boxed{D^{ij,kl}_{F,\mathrm{TT}}(k) = \frac{\Lambda^{ij,kl}(\hat{k})}{k^2 + i\varepsilon}} \tag{3.59}$$

This gauge isolates the radiative degrees of freedom $h^{(+)}_{ij}$ and $h^{(\times)}_{ij}$. Moreover, the angular part of the Fourier transform of its propagator, i.e. the integral of the projector $\Lambda_{ij,kl}(\hat{k})$ over the directions of the momentum, can be expressed in terms of spherical Bessel functions, as we show in Appendix A.

4 Interaction picture

The previous sections were dedicated to the quantization of the free fields. Nevertheless, the most interesting feature of QFT is its ability to make predictions about the interactions between different kinds of particles. Linearized gravity is particularly well suited to this description: its natural Taylor expansion in κ, with matter and gravity terms of decreasing magnitude, makes Feynman diagrams a convenient tool even for computing classical interactions, as we will see. Furthermore, this framework allows one to compute the dominant contributions to the scattering of gravitons and matter particles at the lowest orders in the velocity, and to identify which physical mechanisms are responsible for the different types of gravitational interaction.

4.1 The scattering matrix

In quantum field theory, the scattering matrix $\hat{S}$ is the central object for computing cross sections. We refer to Greiner [10] for a detailed discussion of the formalism; here we only point out what is needed for our purposes. The key step is to identify which part of the Hamiltonian generates the free evolution and which part describes the interactions. We split the Hamiltonian as

$$\hat{H} = \hat{H}_0^{\mathrm{KG}} + \hat{H}_0^{\mathrm{PF}} + \hat{H}^I + \hat{H}^{SI}, \tag{4.1}$$

where each piece arises from the corresponding Lagrangian term: the free Klein–Gordon part $\hat{H}_0^{\mathrm{KG}}$ from (2.38), the free Pauli–Fierz part $\hat{H}_0^{\mathrm{PF}}$ from (3.2) (see (3.5)), the gravity–matter interaction $\hat{H}^I$ from (2.40) and (2.41), and the gravitational self-interaction $\hat{H}^{SI}$ from (2.39). This notation is only provisional, but it makes immediately clear which terms generate the free evolution and which ones describe the interactions. We refer to (2.38) and (2.37) as the free Lagrangians because they are the leading-order terms and because they generate plane-wave solutions of the EoM. The free Hamiltonian density of a massive scalar field is well known in the literature.

The scattering matrix is built from the interaction-picture Hamiltonian, in which the fields and their conjugate momenta evolve freely. Operators are transformed according to

$$\hat{O}^I(t) = \hat{U}(t)\,\hat{O}^H(t)\,\hat{U}^{-1}(t), \qquad \hat{U}(t) = e^{i\hat{H}_0 t}e^{-i\hat{H}t}, \qquad \hat{H}_0 = \hat{H}_0^{\mathrm{KG}} + \hat{H}_0^{\mathrm{PF}}. \tag{4.2}$$

Remarkably, performing a standard Legendre transformation, we find for the *interaction terms*

$$\hat{\mathcal{H}}^I = -\hat{\mathcal{L}}^I[\Phi^I] + \mathcal{N}, \tag{4.3}$$

where $\mathcal{N}$ is a normal-dependent term, i.e. a term that can be expressed through the unit vector n^μ normal to the space-like hypersurface on which we quantize; for the equal-time condition $x_0 = x_0'$, $n^\mu = (1, \vec{0})$ (Ref. Greiner, Example 8.6 [10]). We develop this subject further in the

Appendix. For the moment we only state the resulting rules [15] (Matthews' theorem):

- use $\hat{\mathcal{H}}^I = -\hat{\mathcal{L}}^I[\Phi^I]$, dropping $\mathcal{N}$;
- use the *modified time-ordered product* T^*, defined so as to commute with all derivatives:

$$T^*\{\partial_{\mu_1}\Phi(x_1)\cdots\partial_{\mu_n}\Phi(x_n)\} \equiv \partial^{x_1}_{\mu_1}\cdots\partial^{x_n}_{\mu_n} T\{\Phi(x_1)\cdots\Phi(x_n)\}. \tag{4.4}$$

We now build the contributions to the scattering matrix, ordered in powers of κ. The Dyson series reads

$$\hat{S} = \hat{U}(+\infty,-\infty) = \sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\int d^4x_1\cdots d^4x_n\; T^*\Big(\hat{\mathcal{H}}^I(x_1)\cdots\hat{\mathcal{H}}^I(x_n)\Big). \tag{4.5}$$

Rather than writing out the interaction densities in full, we label them by the vertex they generate:

$$\hat{\mathcal{H}}^I = \underbrace{\hat{\mathcal{H}}^1_M}_{\phi\phi h,\, O(\kappa)} + \underbrace{\hat{\mathcal{H}}^1_g}_{hhh,\, O(\kappa)} + \underbrace{\hat{\mathcal{H}}^2_M}_{\phi\phi hh,\, O(\kappa^2)} + O(\kappa^3), \tag{4.6}$$

their explicit expressions being those obtained in Section 1. The cubic vertex hhh is the quantum counterpart of the non-linearity of the Einstein equations: classically it corrects the gravitational field, while here it appears as a graviton self-interaction [16, 2].

Note that (4.5) is graded by the *number of vertices*, but we regroup it in powers of κ. Moreover, we only write the terms relevant for the scattering of a massive scalar with gravitons, up to second order:

$$\hat{S}_\kappa = -i\int d^4x\,\Big(:\hat{\mathcal{H}}^1_M(x):+:\hat{\mathcal{H}}^1_g(x):\Big), \tag{4.7}$$

$$\begin{aligned}\hat{S}_{\kappa^2} = &-i\int d^4x\,:\hat{\mathcal{H}}^2_M(x): && \text{(seagull)}\\ &-\frac{1}{2}\int d^4x_1\,d^4x_2\;T^*\big\{\hat{\mathcal{H}}^1_M(x_1)\,\hat{\mathcal{H}}^1_M(x_2)\big\} && \text{(scalar exchange)}\\ &-\int d^4x_1\,d^4x_2\;T^*\big\{\hat{\mathcal{H}}^1_M(x_1)\,\hat{\mathcal{H}}^1_g(x_2)\big\} && \text{(graviton exchange)}.\end{aligned} \tag{4.8}$$

Normal ordering is already applied *within* each vertex, so as to discard self-contractions; the contractions *between* the two vertices generate the internal lines via Wick's theorem.

4.2 Feynman rules

Propagators

$$\text{scalar:}\qquad \langle 0|T^*\phi(x)\phi(y)|0\rangle = \int\frac{d^4k}{(2\pi)^4}\,\frac{i\,e^{-ik(x-y)}}{k^2-m^2+i\epsilon}, \tag{4.9}$$

graviton (de Donder): $$\langle 0|T^* h_{\mu\nu}(x) h_{\alpha\beta}(y)|0\rangle = \int \frac{d^4k}{(2\pi)^4} \frac{i\, G_{\mu\nu,\alpha\beta}}{k^2 + i\epsilon} e^{-ik(x-y)}. \tag{4.10}$$

External legs Scalars contribute a factor 1. As shown by the Gupta–Bleuler analysis, physical gravitons can be TT, so each external graviton contributes a TT polarization tensor $\epsilon^{(\lambda)}_{ij}(\hat{k})$, $\lambda = +, \times$, satisfying

$$\epsilon_{ij} = \epsilon_{ji}, \qquad \delta^{ij}\epsilon_{ij} = 0, \qquad k^i \epsilon_{ij} = 0, \qquad \epsilon^{(\lambda)}_{ij} \epsilon^{(\lambda')\, ij} = \delta^{\lambda\lambda'}. \tag{4.11}$$

Scattering matrix The terms of (4.7) and (4.8) are

$$\hat{S}_\kappa = -\frac{i\kappa}{2} \int d^4x \, :h^{\mu\nu} T_{\mu\nu}: \; + \; i\kappa \int d^4x \, :\mathcal{V}_3: \tag{4.12}$$

$$\begin{aligned} \hat{S}_{\kappa^2} = i\kappa^2 \int d^4x \, :\Big[&\frac{1}{4}\Big(\frac{1}{2}h^2 - h_{\alpha\beta}h^{\alpha\beta}\Big)\Big(\frac{1}{2}\partial_\rho\phi\,\partial^\rho\phi - \frac{1}{2}m^2\phi^2\Big) \\ &- \frac{1}{4} h\, h^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi + \frac{1}{2} h^{\mu\alpha} h^{\nu}_{\alpha}\,\partial_\mu\phi\,\partial_\nu\phi \Big]: \end{aligned} \tag{4.13}$$

$$- \frac{\kappa^2}{8} \int d^4x_1\, d^4x_2 \; T^*\big\{ :h^{\mu\nu}T_{\mu\nu}(x_1): : h^{\rho\sigma}T_{\rho\sigma}(x_2):\big\} \tag{4.14}$$

$$+ \frac{\kappa^2}{2} \int d^4x_1\, d^4x_2 \; T^*\big\{ :h^{\mu\nu}T_{\mu\nu}(x_1): : \mathcal{V}_3(x_2):\big\}, \tag{4.15}$$

Where $\mathcal{V}_3$ is the cubic vertex of the graviton self-interaction (2.39).

5 Gravitational scattering of scalar and gravitational waves

5.1 Introduction to the problem

In this section we compute, within the QFT framework, the scattering of a massless wave by the gravitational field of a static mass M [17, 18]. We consider both a scalar and a gravitational wave, and we work in the *Born approximation*: the field of the source is treated as a classical external background, and only the leading order of the Dyson series is kept. In terms of Feynman diagrams, the wave exchanges a single graviton with the source.

There are some assumptions which define the problem: The background does not depend on time, the energy of the wave is conserved and therefore the scattering is elastic.
The wavelength is long compared with the size of the source, so the wave cannot probe its internal structure, and only the monopole part of the field ($\propto 1/r$) is relevant. The angular momentum of the source is neglected (see [17, 18]).

The situation is analogous to the Born approximation in electromagnetic scattering, where a long-wavelength wave interacts with a small body and the field inside the scatterer is approximated by the incident wave. There is, however, a subtlety. The Born approximation requires the scattering field to be weak, but this does not mean that the source must be light: the approximation also holds for a very massive source, as long as the energy of the scattered wave is small enough,

$$\omega/M \ll 1, \tag{5.1}$$

we will investigate this later.

This computation has a double meaning. On the one hand, it shows that a classical, geometrical result can be obtained with QFT techniques. The weak-field cross sections were derived by Peters by solving the wave equations on the Schwarzschild background (Ref. Peters, Phys. Rev. D **13**, 775 (1976) [4]); for a scalar wave and for an unpolarized gravitational wave they read

$$\left.\frac{d\sigma}{d\Omega}\right|_{s=0} = \frac{G^2M^2}{\sin^4(\theta/2)}, \qquad \left.\frac{d\sigma}{d\Omega}\right|_{s=2} = \frac{G^2M^2}{\sin^4(\theta/2)}\left[\cos^8\frac{\theta}{2} + \sin^8\frac{\theta}{2}\right], \tag{5.2}$$

and we will recover both of them from tree-level diagrams. On the other hand, the same amplitudes describe the scattering of a low-energy scalar or graviton off a very massive particle: they are the static limit of the Compton amplitude computed in Section 5.

The simplest case of a monopole source is the Schwarzschild metric outside a spherical body of mass M [7]:

$$g_{00} = 1 - \frac{2GM}{r}, \qquad g_{0j} = 0, \qquad g_{jk} = -\left(1 + \frac{2GM}{r}\right)\delta_{jk}. \tag{5.3}$$

Comparing with $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$, the background has $h_{00} = h_{11} = h_{22} = h_{33} \equiv \Phi$, with trace $h = -2\Phi$ and

$$\Phi = -\frac{\kappa M}{16\pi r}, \qquad \bar{h}_{00} = 2\Phi = -\frac{\kappa M}{8\pi r}, \qquad \bar{h}_{0j} = 0, \qquad \bar{h}_{jk} = 0. \tag{5.4}$$

Notice that this background satisfies the de Donder condition $\partial^\mu \bar{h}_{\mu\nu} = 0$, since it is static and only $\bar{h}_{00}$ is non-zero.

The background is a *static field*, so we are interested in its spatial Fourier transform:

$$\bar{h}_{00}(\mathbf{q}) = -\frac{\kappa M}{2\,|\mathbf{q}|^2}, \qquad \bar{h}_{0j}(\mathbf{q}) = 0, \qquad \bar{h}_{jk}(\mathbf{q}) = 0, \tag{5.5}$$

and

$$h_{00}(\mathbf{q}) = h_{11}(\mathbf{q}) = h_{22}(\mathbf{q}) = h_{33}(\mathbf{q}) = \Phi(\mathbf{q}) = -\frac{\kappa M}{4\,|\mathbf{q}|^2}. \tag{5.6}$$

5.2 Scalar waves cross section

The process we consider is the absorption, by the scalar, of a graviton with momentum q emitted by the background (Fig. 1).

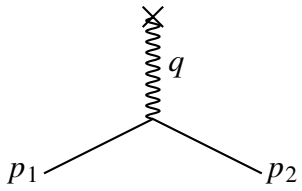


Figure 1: Straight line: scalar; wavy line: graviton

In $\hat{S}_\kappa$, (4.12), we replace the graviton field by the classical background. $T_{\mu\nu}$ combines with $h^{\mu\nu}$, but we can get rid of trace terms in the computations considering $T_{\mu\nu}\bar{h}^{\mu\nu} = \bar{T}_{\mu\nu}h^{\mu\nu}$, we then have:

$$S_{fi} = \langle p_2|\, \hat{S}_\kappa\, |p_1\rangle = -\frac{i\kappa}{2}\int d^4x\; \bar{h}^{\mu\nu}(x)\,\frac{p_{1\mu}p_{2\nu} + p_{2\mu}p_{1\nu}}{(2\pi)^3\sqrt{4\omega\,\omega'}}\; e^{i(p_2-p_1)\cdot x}, \tag{5.7}$$

where ω and ω' are the energies of the incoming and outgoing scalar. $\bar{h}^{\mu\nu} \propto e^{-iq\cdot x}$ sets momentum conservation,

$$p_2 = p_1 + q, \tag{5.8}$$

since the background is static $\omega = \omega'$ and time integration gives $2\pi\delta(\omega' - \omega)$:

$$q^0 = 0, \qquad \mathbf{q} = \mathbf{p}_2 - \mathbf{p}_1, \qquad |\mathbf{p}_2| = |\mathbf{p}_1| = \omega. \tag{5.9}$$

The scattering is therefore elastic, with $|\mathbf{q}| = 2\omega \sin(\theta/2)$, and (5.7) takes the form

$$S_{fi} = -2\pi i\, \delta(\omega' - \omega)\, \frac{\mathcal{M}}{(2\pi)^3\, 2\omega}, \qquad \mathcal{M} = \frac{\kappa}{2}\, \bar{h}^{\mu\nu}(\mathbf{q})\, \big(p_{1\mu} p_{2\nu} + p_{2\mu} p_{1\nu}\big). \tag{5.10}$$

Only $\bar{h}_{00}$ contributes to the vertex:

$$\mathcal{M} = \kappa\, \omega^2\, \bar{h}_{00}(\mathbf{q}) = -\frac{\kappa^2 M \omega^2}{2\,|\mathbf{q}|^2} = -\frac{16\pi G M \omega^2}{|\mathbf{q}|^2}. \tag{5.11}$$

Finally, we resemble for classical cases the differential cross section for such assumptions, $d\sigma/d\Omega = |\mathcal{M}|^2/16\pi^2$ (Ref. Greiner, QED [1]):

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{16\, G^2 M^2 \omega^4}{|\mathbf{q}|^4} = \frac{G^2 M^2}{\sin^4(\theta/2)}.} \tag{5.12}$$

We resembled the result of Peters for a scalar wave, (5.2). It has a Rutherford-type behaviour, $\propto \sin^{-4}(\theta/2)$. Like in the analogous case of QED, it does not depend on ω of the incoming wave, resembling the non dependence on the energy of the incoming particle in the Rutherford scattering.

In the forward limit of small angles, one recovers $d\sigma/d\Omega \to 16G^2M^2/\theta^4$, that's interesting, because it's the same cross section associated with the Schwarzschild deflection of light in geometrical approaches, where $\theta = 4GM/b$. We just saw that for small scattering angles the classical potential $\propto 1/r$ dominates.

Moreover, the backward scattering is finite:

$$\left.\frac{d\sigma}{d\Omega}\right|_{\theta=\pi} = G^2 M^2, \tag{5.13}$$

which is a wave effect, absent in the geometrical-optics description.

5.3 Gravitational waves cross section

The same background also deflects a gravitational wave. In this case the coupling comes from the cubic self-interaction of gravity, and the amplitude is obtained by taking one leg of $\mathcal{V}_3$ to be the background and the other two to be the incoming and outgoing gravitons.

The incoming and outgoing gravitons are set to

$$h^{(1)}_{\mu\nu} \propto \epsilon^{(1)}_{\mu\nu}\, e^{-ik_1\cdot x}, \qquad h^{(2)}_{\mu\nu} \propto \epsilon^{(2)}_{\mu\nu}\, e^{+ik_2\cdot x}, \tag{5.14}$$

with TT polarizations. We repeat the same computing as before, writing $\epsilon^{(2)}\cdot\,\epsilon^{(1)} \equiv \epsilon^{(2)}_{ij}\epsilon^{(1)}_{ij}$,

each of the contributions from $\mathcal{V}_3$ is:

$$\begin{aligned}
\tfrac{1}{2} h^\alpha_\beta \partial^\mu h^\beta_\alpha \partial_\mu h &\to +\Phi\,|\mathbf{q}|^2\,(\epsilon^{(2)}\cdot\epsilon^{(1)}),\\
-\tfrac{1}{2} h^\alpha_\beta \partial_\alpha h^\mu_\nu \partial^\beta h^\nu_\mu &\to -\Phi\,\omega^2(1+\cos\theta)\,(\epsilon^{(2)}\cdot\epsilon^{(1)}),\\
-h^\alpha_\beta \partial_\mu h^\nu_\alpha \partial^\mu h^\beta_\nu &\to -\Phi\,|\mathbf{q}|^2\,(\epsilon^{(2)}\cdot\epsilon^{(1)}),\\
\tfrac{1}{4} h\,\partial^\beta h^\mu_\nu \partial_\beta h^\nu_\mu &\to -\tfrac{1}{2}\Phi\,|\mathbf{q}|^2\,(\epsilon^{(2)}\cdot\epsilon^{(1)}),\\
h^\beta_\mu \partial_\nu h^\alpha_\beta \partial^\mu h^\nu_\alpha &\to 0,\\
-\tfrac{1}{8} h\,\partial^\nu h\,\partial_\nu h &\to 0.
\end{aligned}$$

Therefore, with the same $|\mathbf{q}|$ as before we have a total

$$\mathcal{M}' = 2\kappa\,\omega^2\,\Phi(\mathbf{q})\,\left(\epsilon^{(2)}\cdot\epsilon^{(1)}\right) = \mathcal{M}\,\left(\epsilon^{(2)}\cdot\epsilon^{(1)}\right), \tag{5.15}$$

resulting in

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{G^2M^2}{\sin^4(\theta/2)}\left|\epsilon^{(2)}\cdot\epsilon^{(1)}\right|^2.} \tag{5.16}$$

The gravitational wave is therefore deflected exactly as the scalar wave, up to a polarization factor. As in the scalar case, the scattering is Rutherford-like. In the forward limit $(\epsilon^{(2)}\cdot\epsilon^{(1)}) \to 1$, for equal polarizations, so we recover exactly the scalar result (5.12).

For an unpolarized wave we average over the initial polarizations and sum over the final ones $\sum_\lambda \epsilon^{(\lambda)}_{ij}\epsilon^{(\lambda)}_{kl}$. We will dedicate to polarization tensors product a section later, for now let us show the cross-section:

$$\boxed{\left.\frac{d\sigma}{d\Omega}\right|_{\text{unpol}} = \frac{G^2M^2}{\sin^4(\theta/2)}\left[\cos^8\frac{\theta}{2} + \sin^8\frac{\theta}{2}\right]} \tag{5.17}$$

which is exactly the result of Peters for gravitational waves, (5.2).

The polarization factor has been studied in a more general case by Guadagnini [18]: precise study could distinguish the polarization of the scattered gravitational wave even in the monopole potential case.

In conclusion, in the Born approximation scalar and gravitational waves are deflected by a static mass in the same way, up to a polarization factor. In the forward limit, where the exchanged graviton is soft ($q \to 0$) and the polarization factor tends to one, we recovered of the equivalence principle for which a soft graviton interaction doesn't depend on its type [15].

6 Graviton–Scalar Compton Scattering

6.1 Introduction to the problem

Now we are interested in calculating the leading order of a Compton-like scattering process [19, 14, 9]

$$\phi(p_1) + h(k_1, \lambda_1) \longrightarrow \phi(p_2) + h(k_2, \lambda_2), \tag{6.1}$$

with states labelled

$$|i\rangle = |p_1; k_1, \lambda_1\rangle\,, \qquad \langle f| = \langle p_2; k_2, \lambda_2|\,. \tag{6.2}$$

We are interested in how the amplitude depends on the velocity of the scalar particle, and in which physical processes are responsible for the different contributions to the scattering.

It is convenient to factor out once and for all

$$\mathcal{N} \equiv \frac{\delta^{(4)}\big(k_{\rm in} - k_{\rm out}\big)}{(2\pi)^2\sqrt{16\,E_1E_2\omega_1\omega_2}}, \qquad k_{\rm in} = p_1 + k_1, \quad k_{\rm out} = p_2 + k_2, \tag{6.3}$$

where E_1, E_2 are the scalar energies and ω_1, ω_2 the graviton ones. Each contribution of $\hat{S}$ will be written in the form

$$S_{fi}^{(\alpha)} = i\,\kappa^2\,\mathcal{N}\,\mathcal{M}^{(\alpha)}, \tag{6.4}$$

where (α) labels the diagram we are considering. In particular, there are four diagrams at $O(\kappa^2)$, as we have seen in Fig. **??**: the seagull, the s- and u-channel scalar exchange, and the t-channel graviton exchange. We compute each of them separately and explicitly in Appendix D; for now, let us show the results.

A common procedure for all the diagrams is to first use momentum conservation and the on-shell conditions for the states, and then go to the classical limit ($\omega \ll m$), where

$$p_1 = p_2 \simeq E\,(1, \vec{v}), \qquad k_{1,2} = \omega\,(1, \vec{n}_{1,2}). \tag{6.5}$$

The **seagull channel**, represented by

$$S_{fi}^{\rm sg} = -i\int d^4x\ \langle p_2; k_2, \lambda_2|\,:\,\hat{\mathcal{H}}_M^2(x)\,:\,|p_1; k_1, \lambda_1\rangle\,, \tag{6.6}$$

can be expressed in terms of a Feynman diagram:

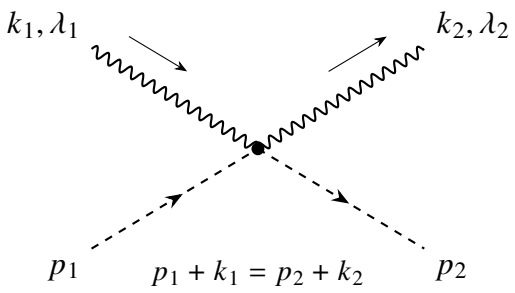


Figure 2: Seagull channel from $\hat{\mathcal{H}}^2_M$.

It comes from the first-order term of the Dyson series, (4.13). We have

$$\mathcal{M}^{\rm sg} = -\frac{\epsilon^{(2)}\cdot\epsilon^{(1)}}{2}\left(p_2\cdot p_1 - m^2\right) + \epsilon^{(2)\,\mu\alpha}\,\epsilon^{(1)\,\nu}{}_{\alpha}\left(p_{2\mu}p_{1\nu} + p_{1\mu}p_{2\nu}\right). \tag{6.7}$$

Since TT polarizations are purely spatial, the expression at lowest order in $\vec{v}$ is

$$\mathcal{M}^{\rm sg} = O(v^2), \tag{6.8}$$

so it contributes neither to the static limit nor to the first order in $\vec{v}$.

Now we derive the **virtual scalar exchange** contributions, represented by the s- and u-channel diagrams:

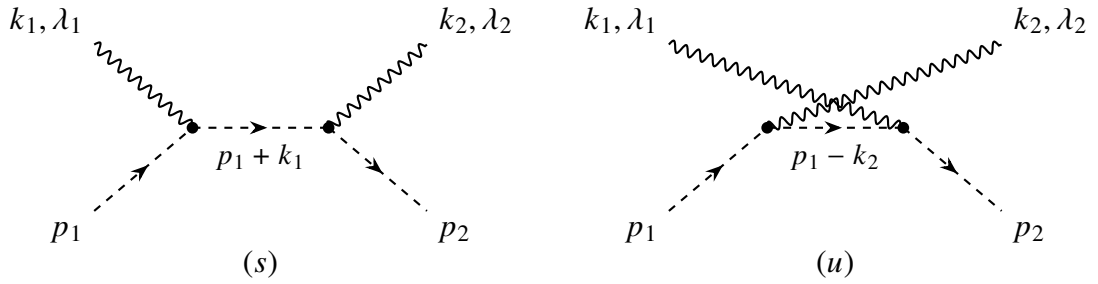


Figure 3: Virtual scalar exchange: s channel (left), u channel (right).

To have a virtual scalar exchange, we have to take two vertices from the scattering matrix:

$$S^{s+u}_{fi} = -\frac{1}{2}\int d^4x_1\, d^4x_2\ \langle p_2; k_2, \lambda_2|\, T^*\left\{:\hat{\mathcal{H}}^1_M(x_1)::\hat{\mathcal{H}}^1_M(x_2):\right\}|p_1; k_1, \lambda_1\rangle\,, \tag{6.9}$$

with the two internal ϕ fields contracted into the Feynman propagator. We obtain

$$\mathcal{M}^s = -\frac{1}{2p_1\cdot k_1 + i\varepsilon}\left(p_1\cdot\epsilon^{(1)}\cdot p_1\right)\left(p_2\cdot\epsilon^{(2)}\cdot p_2\right), \tag{6.10}$$

$$\mathcal{M}^u = +\frac{1}{2p_1\cdot k_2 + i\varepsilon}\left(p_1\cdot\epsilon^{(2)}\cdot p_1\right)\left(p_2\cdot\epsilon^{(1)}\cdot p_2\right). \tag{6.11}$$

If we now take the classical limit again, we see that

$$\mathcal{M}^s + \mathcal{M}^u \simeq O(v^4). \tag{6.12}$$

The scalar exchange is $O(v^4)$: it does not contribute to the static limit and is of higher order in v than the seagull term.

This is interesting because, as we will see, the only remaining contribution gives a Rutherford-like scattering; therefore, we have found that the leading-order non-Rutherford cross section is $O(v^2)$.

Finally, we compute the leading-order contribution in v to the Compton process, the **graviton self-interaction channel**. As usual, we draw it as a Feynman diagram, with the exchange of an internal graviton:

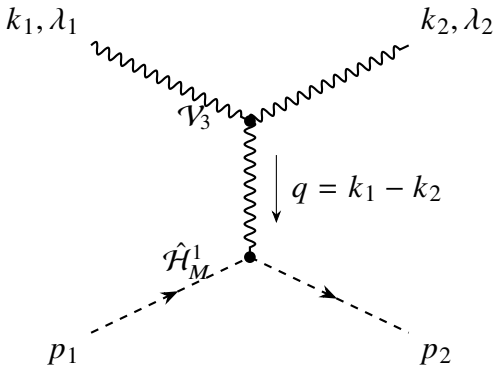


Figure 4: t-channel.

The cubic self-interaction $\mathcal{V}_3$ has a complicated structure, but since all its legs are in the de Donder gauge (two are TT external gravitons and one is contracted with the de Donder propagator), we can use the expression we wrote in Sec. 5:

$$\begin{aligned}\mathcal{V}_3 = &\underbrace{\tfrac{1}{2} h^\alpha_\beta \partial^\mu h^\beta_\alpha \partial_\mu h}_{(1)} \underbrace{-\tfrac{1}{2} h^\alpha_\beta \partial_\alpha h^\mu_\nu \partial^\beta h^\nu_\mu}_{(2)} \underbrace{-h^\alpha_\beta \partial_\mu h^\nu_\alpha \partial^\mu h^\beta_\nu}_{(3)} \\ &+ \underbrace{\tfrac{1}{4} h\, \partial^\beta h^\mu_\nu \partial_\beta h^\nu_\mu}_{(4)} + \underbrace{h^\beta_\mu \partial_\nu h^\alpha_\beta \partial^\mu h^\nu_\alpha}_{(5)} \underbrace{-\tfrac{1}{8} h\, \partial^\nu h\, \partial_\nu h}_{(6)} .\end{aligned} \tag{6.13}$$

In terms of the Dyson series, from (4.15) we have

$$S^t_{fi} = \frac{\kappa^2}{2} \int d^4x_1\, d^4x_2\ \langle p_2; k_2, \lambda_2|\ T^* \left\{ : h^{\mu\nu} T_{\mu\nu}(x_1) : : \mathcal{V}_3(x_2) : \right\}\ |p_1; k_1, \lambda_1\rangle\,, \tag{6.14}$$

where we have to consider all possible contractions. These computations, which we report

directly in the classical limit, give the following result:

$$S^t_{fi} \simeq i\kappa^2 \mathcal{N} \frac{E^2}{4\sin^2(\theta/2)} \mathcal{M} = \frac{i\,GE}{2\pi\,\omega\,\sin^2(\theta/2)} \delta^{(4)}\big(k_{\rm in} - k_{\rm out}\big)\,\mathcal{M}, \tag{6.15}$$

where, in powers of the velocity, $\mathcal{M} = \mathcal{M}^{(0)} + \mathcal{M}^{(1)} + O(v^2)$, with

$$\mathcal{M}^{(0)} = (\epsilon^{(2)}\cdot\epsilon^{(1)}), \tag{6.16}$$

$$\mathcal{M}^{(1)} = -(\epsilon^{(2)}\cdot\epsilon^{(1)})\,\vec{v}\cdot(\vec{n}_2+\vec{n}_1) + 2\big(\vec{v}\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot\vec{n}_2 + \vec{v}\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot\vec{n}_1\big). \tag{6.17}$$

The static term $\mathcal{M}^{(0)}$ is the polarization factor found for the deflection of a gravitational wave by a fixed mass, Eq. (5.15).

The first-order term $\mathcal{M}^{(1)}$ describes how the motion of the mass modifies the scattering.

7 Scattering cross section

We compute the scattering cross section in the classical limit discussed above, together with its first-order correction in the scalar velocity $\vec{v}$. We use the real TT polarizations $\lambda = +, \times$ from Section 1. First of all, we make some considerations on the products of spin-2 polarization tensors.

7.1 Polarization products

To compute spin-2 polarization tensor products, we introduce a generalization of the spin-1 case treated in Greiner, *QED* [1].

The TT polarization tensors live in the plane transverse to the graviton momentum. For each momentum $\vec{k}_i$, we introduce two real orthonormal vectors $\vec{u}_i, \vec{w}_i$ in that plane,

$$\vec{u}_i \cdot \vec{w}_i = 0, \qquad \vec{u}_i^{\,2} = \vec{w}_i^{\,2} = 1,$$

and write

$$\epsilon_i^{(+)} = \tfrac{1}{\sqrt{2}}\big(\vec{u}_i\,\vec{u}_i - \vec{w}_i\,\vec{w}_i\big), \qquad \epsilon_i^{(\times)} = \tfrac{1}{\sqrt{2}}\big(\vec{u}_i\,\vec{w}_i + \vec{w}_i\,\vec{u}_i\big).$$

That's coherent for the physical modes we found in de Donder gauge. We choose the bases so that $\vec{u}_1, \vec{u}_2$ lie in the scattering plane spanned by $\vec{k}_1, \vec{k}_2$, and $\vec{w}_1 = \vec{w}_2$ is orthogonal to it, as shown in Fig. 5.

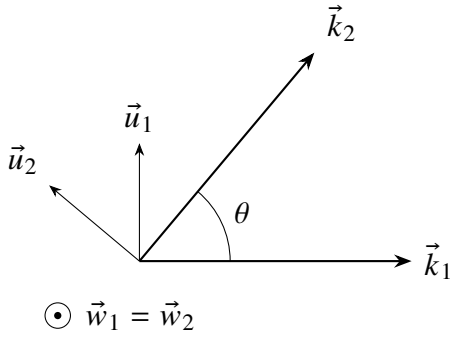


Figure 5: Polarization basis. The vectors $\vec{u}_1, \vec{u}_2$ lie in the scattering plane, while $\vec{w}_1 = \vec{w}_2$ points out of it.

Then

$$\vec{u}_1 \cdot \vec{u}_2 = \cos\theta, \qquad \vec{w}_1 \cdot \vec{w}_2 = 1, \qquad \vec{u}_1 \cdot \vec{w}_2 = \vec{w}_1 \cdot \vec{u}_2 = 0. \tag{7.1}$$

The polarization products are

$$\epsilon(\vec{k}_2, +) \cdot \epsilon(\vec{k}_1, +) = \tfrac{1}{2}\big(1 + \cos^2\theta\big), \tag{7.2}$$

$$\epsilon(\vec{k}_2,+)\cdot\epsilon(\vec{k}_1,\times)=\epsilon(\vec{k}_2,\times)\cdot\epsilon(\vec{k}_1,+)=0, \tag{7.3}$$

$$\epsilon(\vec{k}_2,\times)\cdot\epsilon(\vec{k}_1,\times)=\cos\theta. \tag{7.4}$$

Averaging over the initial and summing over the final polarizations,

$$\boxed{\frac{1}{2}\sum_{\lambda_1,\lambda_2=+,\times}\left|\epsilon^{(2)}\cdot\epsilon^{(1)}\right|^2=\frac{1+6\cos^2\theta+\cos^4\theta}{8}=\cos^8\tfrac{\theta}{2}+\sin^8\tfrac{\theta}{2}} \tag{7.5}$$

The first-order scattering terms contain $\vec{v}\cdot\vec{n}_i$ and $\vec{v}\cdot\epsilon^{(i)}\cdot\epsilon^{(j)}\cdot\vec{n}_i$, which depend on the direction of $\vec{v}$. Let's fix an initial state of a *scalar moving in the same direction of the graviton*, $\vec{v}=v\,\vec{n}_1=v\,\hat{z}$. Then

$$\vec{v}\cdot\vec{n}_1=v,\qquad \vec{v}\cdot\vec{n}_2=v\cos\theta. \tag{7.6}$$

Transversality gives

$$\vec{v}\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot\vec{n}_1=0, \tag{7.7}$$

therefore, for the polarizations (λ_1,λ_2),

$$\vec{v}\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot\vec{n}_2=v\left(\vec{n}_1\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot\vec{n}_2\right)=\begin{cases}-\frac{1}{2}\,v\cos\theta\,\sin^2\theta & (+,+),\\ -\frac{1}{2}\,v\,\sin^2\theta & (\times,\times),\\ 0 & (+,\times),\ (\times,+).\end{cases} \tag{7.8}$$

7.2 Classical limit cross section

Up to first order in $\vec{v}$, only the t-channel contributes, since the seagull channel is $O(v^2)$ and the s,u channels are $O(v^4)$. From (6.15),

$$S^t_{fi}=i\kappa^2\,\mathcal{N}\,\frac{E^2}{4\sin^2(\theta/2)}\,\mathcal{M}. \tag{7.9}$$

For the scalar at rest, $\mathcal{M}\simeq\mathcal{M}^{(0)}$.

Following Greiner [10, 1], we regularize the delta functions in a continuous volume $V\to(2\pi)^3$ and time T, with the usual substitution

$$\left[\delta^{(4)}(k_{\text{in}}-k_{\text{out}})\right]^2=\frac{T}{2\pi}\,\delta^{(4)}(k_{\text{in}}-k_{\text{out}}). \tag{7.10}$$

Hence

$$|S^t_{fi}|^2=\frac{\kappa^4E^4|\mathcal{M}|^2}{16\sin^4(\theta/2)}\,\frac{T\,\delta^{(4)}(k_{\text{in}}-k_{\text{out}})}{(2\pi)^5\,16E_1E_2\omega_1\omega_2}. \tag{7.11}$$

We divide by the time T, by the incident flux $|\vec{v}_{\rm rel}|/V$ and by normalization factors:

$$d\sigma = \frac{V}{T\,|\vec{v}_{\rm rel}|}\,|S^t_{fi}|^2\,d^3p_2\,d^3k_2 = \frac{\kappa^4 E^4 |\mathcal{M}|^2}{16\sin^4(\theta/2)}\,\frac{\delta^{(4)}(k_{\rm in}-k_{\rm out})}{(2\pi)^2\,16E_1E_2\omega_1\omega_2\,|\vec{v}_{\rm rel}|}\,d^3p_2\,d^3k_2. \quad (7.12)$$

For the scalar at rest, in the classical limit we have

$$\boxed{\left.\frac{d\sigma}{d\Omega}\right|_{v=0} = \frac{G^2E^2}{\sin^4(\theta/2)}\left|\epsilon^{(2)}\cdot\epsilon^{(1)}\right|^2,} \quad (7.13)$$

which again is Rutherford-like. For the unpolarized case we then have poved to have the same result as Peters [4]:

$$\left.\frac{d\sigma}{d\Omega}\right|^{\rm unpol}_{v=0} = \frac{G^2E^2}{\sin^4(\theta/2)}\left(\cos^8\tfrac{\theta}{2} + \sin^8\tfrac{\theta}{2}\right). \quad (7.14)$$

7.3 Correction at linear order in v

Now we evaluate the first-order quantum-field correction in the velocity. Since $E = m + O(v^2)$, the corrections come from the following elements of (7.12):

- energy conservation: from $p_1\cdot k_1 = p_2\cdot k_2$ we recover the elastic scattering condition

$$\omega_2 = \omega_1\,\frac{1-\vec{n}_1\cdot\vec{v}}{1-\vec{n}_2\cdot\vec{v}}; \quad (7.15)$$

- delta function factors: since E_2 depends on ω_2, $\int d\omega_2\,\delta(E_1+\omega_1-E_2-\omega_2)$ gives a factor $(1-\vec{n}_2\cdot\vec{v})^{-1}$;

- flux: $|\vec{v}_{\rm rel}| = 1-\vec{n}_1\cdot\vec{v}$.

Inserting these into the general form (7.12), we obtain

$$\frac{d\sigma}{d\Omega} = \frac{G^2E^2}{\sin^4(\theta/2)}\,\frac{|\mathcal{M}|^2}{(1-\vec{n}_2\cdot\vec{v})^2}. \quad (7.16)$$

Then we consider also scattering amplitudes corrections, $|\mathcal{M}|^2 = |\mathcal{M}^{(0)}|^2 + 2\,\mathcal{M}^{(0)}\mathcal{M}^{(1)} + O(v^2)$:

$$\frac{d\sigma}{d\Omega} = \frac{G^2E^2}{\sin^4(\theta/2)}\Big[\,|\mathcal{M}^{(0)}|^2\big(1+2\,\vec{n}_2\cdot\vec{v}\big) + 2\,\mathcal{M}^{(0)}\mathcal{M}^{(1)}\Big]. \quad (7.17)$$

Finally, we have a formule which reduces to (7.13) when $v = 0$, which is consistent and which can be explicited in:

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{G^2E^2}{\sin^4(\theta/2)}\Big\{\big(\epsilon^{(2)}\cdot\epsilon^{(1)}\big)^2\big(1-2\,\vec{n}_1\cdot\vec{v}\big) + 4\,\big(\epsilon^{(2)}\cdot\epsilon^{(1)}\big)\big(\vec{v}\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot\vec{n}_2+\vec{v}\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot\vec{n}_1\big)\Big\} + O(v^2).} \tag{7.18}$$

We could make it explicit for the particular cases of the collinear graviton-scalar scattering, using the polarization products computed above, however we are more interested in noticing that there are two types of corrections:

- a reduction of the cross section due to the relative velocity: the scalar sees a redshifted graviton and, since gravity couples to energy, it is less likely to scatter it;
- an angular contribution, coming from the tensorial structure of the graviton, which vanishes for $\theta = n\,\pi/2$.

8 Conclusions??

In this thesis we have treated linearized gravity as a quantum field theory and applied it to the scattering of gravitons off scalar matter.

Starting from the Einstein–Hilbert action, we derived the free Lagrangians and the interaction vertices up to $O(\kappa^2)$. We then quantized the graviton canonically in two gauges. In the de Donder gauge we quantized the metric perturbation $h_{\mu\nu}$ itself, rather than its trace-reversed counterpart: the commutation relations are no longer diagonal in the polarization index, but the propagator coincides with the one obtained by Gupta [12] and from the path integral, and the Gupta–Bleuler condition selects physical states that are transverse and traceless. In the TT gauge, instead, only the two physical polarizations are quantized, at the price of losing manifest covariance.

With these tools we computed the scattering of scalar and gravitational waves by a static mass. The tree-level diagrams reproduce exactly the classical cross sections of Peters [4]: both waves undergo a Rutherford-like scattering, and they differ only by a polarization-dependent factor.

Finally, we studied graviton–scalar Compton scattering, expanding the amplitude in the velocity of the scalar. Up to first order only the graviton-exchange diagram contributes, while the seagull and scalar-exchange diagrams are of higher order. The static limit recovers the fixed-background result, and the first-order correction consists of a Doppler factor and of an angular term due to the spin-2 nature of the graviton.

These results can be extended in several directions: higher orders in the velocity, where the remaining diagrams become relevant; sources with angular momentum; and the description of gravity-induced decoherence within the influence-functional formalism [5, 3], for which the propagators derived here are the starting point.

A Bessel expression for the TT-gauge propagator

In the open-quantum-system formalism [5, 20] one ofter needs the graviton propagator in position space. For the TT gauge, the Fourier transform of the propagator contains the angular integral of the projector $\Lambda_{ij,kl}$ over the direction $\hat{n}$ of the momentum $\vec{k} = k\,\hat{n}$,

$$I_{ij,kl}(\vec{r}, k) \equiv \int d\Omega_{\hat{n}}\; \Lambda_{ij,kl}(\hat{n})\, e^{ik\hat{n}\cdot\vec{r}}, \qquad \vec{r} = \vec{x} - \vec{y}. \tag{A.1}$$

Here we show that this integral can be expressed in terms of spherical Bessel functions.

A.1 Reduction to derivatives of a scalar integral

Expanding the products in (2.35), the projector is a sum of terms of degree 0, 2 and 4 in $\hat{n}$:

$$2\,\Lambda_{ij,kl}(\hat{n}) = D^{(0)}_{ijkl} - D^{(2)}_{ijkl}(\hat{n}) + \hat{n}_i\hat{n}_j\hat{n}_k\hat{n}_l, \tag{A.2}$$

with

$$D^{(0)}_{ijkl} = \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \delta_{ij}\delta_{kl}, \tag{A.3}$$

$$D^{(2)}_{ijkl}(\hat{n}) = \delta_{ik}\hat{n}_j\hat{n}_l + \delta_{jl}\hat{n}_i\hat{n}_k + \delta_{il}\hat{n}_j\hat{n}_k + \delta_{jk}\hat{n}_i\hat{n}_l - \delta_{ij}\hat{n}_k\hat{n}_l - \delta_{kl}\hat{n}_i\hat{n}_j. \tag{A.4}$$

The integral (A.1) therefore reduces to three integrals,

$$J^{(0)} \equiv \int d\Omega\; e^{ik\hat{n}\cdot\vec{r}}, \qquad J^{(2)}_{ab} \equiv \int d\Omega\; \hat{n}_a\hat{n}_b\, e^{ik\hat{n}\cdot\vec{r}}, \qquad J^{(4)}_{abcd} \equiv \int d\Omega\; \hat{n}_a\hat{n}_b\hat{n}_c\hat{n}_d\, e^{ik\hat{n}\cdot\vec{r}}. \tag{A.5}$$

Each factor $\hat{n}_a$ can be replaced by a derivative with respect to $\vec{r}$,

$$\hat{n}_a\, e^{ik\hat{n}\cdot\vec{r}} = \frac{1}{ik}\,\partial_a\, e^{ik\hat{n}\cdot\vec{r}}, \qquad \partial_a \equiv \frac{\partial}{\partial r_a}, \tag{A.6}$$

which can be taken out of the integral. Hence

$$J^{(2)}_{ab} = -\frac{1}{k^2}\,\partial_a\partial_b\, J^{(0)}, \qquad J^{(4)}_{abcd} = \frac{1}{k^4}\,\partial_a\partial_b\partial_c\partial_d\, J^{(0)}, \tag{A.7}$$

and everything follows from the scalar integral

$$J^{(0)}(\vec{r}, k) = 4\pi\,\frac{\sin(kr)}{kr} = 4\pi\, j_0(x), \qquad r = |\vec{r}|, \quad x \equiv kr, \tag{A.8}$$

where j_0 is the zeroth-order spherical Bessel function.

A.2 Derivatives of a radial function

For a function $f(r)$ depending on $\vec{r}$ only through r, we have $\partial_a f = r_a\, \mathcal{D} f$, where

$$\mathcal{D} \equiv \frac{1}{r}\frac{d}{dr}. \tag{A.9}$$

Using $\partial_a r_b = \delta_{ab}$ and applying this rule repeatedly, we find

$$\partial_a \partial_b\, f = \delta_{ab}\, \mathcal{D} f + r^2\, \hat{r}_a \hat{r}_b\, \mathcal{D}^2 f, \tag{A.10}$$

$$\partial_a \partial_b \partial_c \partial_d\, f = T^{(1)}_{abcd}\, \mathcal{D}^2 f + r^2\, T^{(2)}_{abcd}\, \mathcal{D}^3 f + r^4\, T^{(3)}_{abcd}\, \mathcal{D}^4 f, \tag{A.11}$$

where $\hat{r} = \vec{r}/r$ and

$$T^{(1)}_{abcd} = \delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}, \tag{A.12}$$

$$T^{(2)}_{abcd} = \delta_{ab}\hat{r}_c\hat{r}_d + \delta_{ac}\hat{r}_b\hat{r}_d + \delta_{ad}\hat{r}_b\hat{r}_c + \delta_{bc}\hat{r}_a\hat{r}_d + \delta_{bd}\hat{r}_a\hat{r}_c + \delta_{cd}\hat{r}_a\hat{r}_b, \tag{A.13}$$

$$T^{(3)}_{abcd} = \hat{r}_a\hat{r}_b\hat{r}_c\hat{r}_d. \tag{A.14}$$

A.3 Spherical Bessel functions

For $f(r) = j_0(kr)$ the operator $\mathcal{D}$ acts in a particularly simple way. The spherical Bessel functions satisfy

$$\left(\frac{1}{x}\frac{d}{dx}\right)^n j_0(x) = (-1)^n\, \frac{j_n(x)}{x^n}, \tag{A.15}$$

and since $\mathcal{D} = k^2\, \frac{1}{x}\frac{d}{dx}$,

$$\mathcal{D}^n j_0(kr) = (-1)^n\, k^{2n}\, \frac{j_n(x)}{x^n}. \tag{A.16}$$

Each application of $\mathcal{D}$ simply raises the order of the Bessel function by one. Inserting (A.16) into (A.10) and (A.11), and using (A.7) and (A.8), we obtain

$$J^{(2)}_{ab} = 4\pi\left[\frac{j_1(x)}{x}\,\delta_{ab} - j_2(x)\,\hat{r}_a\hat{r}_b\right], \tag{A.17}$$

$$J^{(4)}_{abcd} = 4\pi\left[\frac{j_2(x)}{x^2}\,T^{(1)}_{abcd} - \frac{j_3(x)}{x}\,T^{(2)}_{abcd} + j_4(x)\,T^{(3)}_{abcd}\right]. \tag{A.18}$$

A.4 Result

Substituting (A.8), (A.17) and (A.18) into (A.2), the angular integral of the TT projector is

$$\boxed{I_{ij,kl}(\vec{r},k) = 2\pi\left[\left(j_0 - \frac{2j_1}{x}\right)D^{(0)}_{ijkl} + j_2\, D^{(2)}_{ijkl}(\hat{r}) + \frac{j_2}{x^2}\,T^{(1)}_{ijkl} - \frac{j_3}{x}\,T^{(2)}_{ijkl} + j_4\,T^{(3)}_{ijkl}\right]} \tag{A.19}$$

B Interaction Hamiltonian terms

When the interaction contains derivatives of the fields, the interaction Hamiltonian is not simply minus the interaction Lagrangian: as stated in (4.3), it contains an additional normal-dependent term $\mathcal{N}$. A naive computation of the scattering matrix would therefore give Feynman rules different from the ones read off directly from the Lagrangian. However, as in scalar QED, these extra terms cancel out. The reason lies in the time derivatives of the two-point function.

A single time derivative produces no extra term, since the fields commute at equal times, $[\Phi(\vec{x},t),\Phi(\vec{y},t)]=0$:

$$\partial_0^x\, T\{\phi^I(x)\phi^I(y)\} = T\{\dot{\phi}^I(x)\phi^I(y)\}. \tag{B.1}$$

With two time derivatives, instead, the equal-time commutator between the field and its conjugate momentum gives a contact term:

$$\partial_0^x\partial_0^y\, T\{\phi^I(x)\phi^I(y)\} = T\{\dot{\phi}^I(x)\dot{\phi}^I(y)\} + i\,\delta^{(4)}(x-y). \tag{B.2}$$

The same happens for the graviton field, where the contact term carries the tensor structure of the commutation relations (3.15):

$$\partial_0^x\partial_0^y\, T\{h^I_{\mu\nu}(x)h^I_{\alpha\beta}(y)\} = T\{\dot{h}^I_{\mu\nu}(x)\dot{h}^I_{\alpha\beta}(y)\} + i\,G_{\mu\nu\alpha\beta}\,\delta^{(4)}(x-y). \tag{B.3}$$

The ordinary time-ordered product T and the product T^* defined in (4.4), in which the derivatives are taken outside the time ordering, therefore differ by contact terms. These contact terms are exactly what cancels the normal-dependent term $\mathcal{N}$ of the Hamiltonian.

A general proof proceeds order by order: every Feynman diagram of a given order contains subgraphs in which two time derivatives act on the same propagator, and the resulting contact terms cancel the contributions of $\mathcal{N}$ at the same order. Since this is tedious to show explicitly in the gravitational case, we stop at the analogy with scalar QED and refer to Greiner (*Field Quantization*, Chap. 8 [10]) and to Matthews' theorem. The result is that the scattering matrix can be computed with the two rules stated in Section 3: use $\hat{\mathcal{H}}^I = -\hat{\mathcal{L}}^I$, dropping $\mathcal{N}$, and replace the time-ordered product T with T^*.

C The states $\hat{L}^\dagger_{k\nu}|0\rangle$

Since $\hat{L}_{k\nu}|0\rangle = 0$, the norm of $\hat{L}^\dagger_{k\nu}|0\rangle$ is given by

$$\langle 0|\,\hat{L}_{k\nu}\hat{L}^\dagger_{k\nu}\,|0\rangle = \langle 0|\,[\hat{L}_{k\nu},\hat{L}^\dagger_{k\nu}]\,|0\rangle\,. \tag{C.1}$$

The only non-zero components of $\mathcal{G}_{\lambda\lambda'}$ are

$$\mathcal{G}_{(a,a)(b,b)} = \delta_{ab} - \tfrac{1}{2}\eta_{aa}\eta_{bb}, \qquad \mathcal{G}_{(0i)(0i)} = -1, \qquad \mathcal{G}_{(ij)(ij)} = +1 \quad (i<j). \tag{C.2}$$

More explicitly, $\mathcal{G}_{(a,a)(a,a)} = \frac{1}{2}$, $\mathcal{G}_{(0,0)(i,i)} = +\frac{1}{2}$ and $\mathcal{G}_{(i,i)(j,j)} = -\frac{1}{2}$ for $i \neq j$. For the four combinations in (3.39) we find

$$[\hat{a}_{01} - \hat{a}_{13},\, \text{h.c.}] = \mathcal{G}_{(01)(01)} + \mathcal{G}_{(13)(13)} = -1 + 1 = 0, \tag{C.3}$$

$$[\hat{a}_{02} - \hat{a}_{23},\, \text{h.c.}] = \mathcal{G}_{(02)(02)} + \mathcal{G}_{(23)(23)} = -1 + 1 = 0, \tag{C.4}$$

$$[\hat{a}_{11} + \hat{a}_{22},\, \text{h.c.}] = \tfrac{1}{2} + \tfrac{1}{2} + 2\big(-\tfrac{1}{2}\big) = 0, \tag{C.5}$$

$$[\hat{a}_{00} - \sqrt{2}\,\hat{a}_{03} + \hat{a}_{33},\, \text{h.c.}] = \tfrac{1}{2} + \tfrac{1}{2} + 2\big(\tfrac{1}{2}\big) + 2\,(-1) = 0. \tag{C.6}$$

The mutual commutators vanish as well. The first two combinations contain only off-diagonal modes, which are orthogonal to each other and to every diagonal mode, so their commutators vanish trivially. For the other two,

$$[\hat{a}_{11}+\hat{a}_{22},\, \hat{a}^\dagger_{00} - \sqrt{2}\,\hat{a}^\dagger_{03} + \hat{a}^\dagger_{33}] = \mathcal{G}_{(11)(00)} + \mathcal{G}_{(11)(33)} + \mathcal{G}_{(22)(00)} + \mathcal{G}_{(22)(33)} = \tfrac{1}{2} - \tfrac{1}{2} + \tfrac{1}{2} - \tfrac{1}{2} = 0. \tag{C.7}$$

Hence $[\hat{L}_{k\mu}, \hat{L}^\dagger_{k\nu}] = 0$ for every μ, ν, and each $\hat{L}^\dagger_{k\nu}\,|0\rangle$ has zero norm.

D Computation of the Compton diagrams

In this appendix we show the computations leading to the results of Section 6, for each of the diagrams at $O(\kappa^2)$.

D.1 Seagull channel

Inserting the mode expansions into S^{sg}_{fi}, after some operator algebra we obtain

$$\begin{aligned} S^{\text{sg}}_{fi} = i\kappa^2 \int d^4x \int \frac{d^3q_1\, d^3q_2\, d^3q_3\, d^3q_4}{(2\pi)^6\sqrt{16 E_{q_1}E_{q_2}E_{q_3}E_{q_4}}} \sum_{\lambda,\lambda'} e^{i(q_1+q_2-q_3-q_4)\cdot x} \\ \times \Big\{ -\tfrac{1}{8}\,\epsilon_{\alpha\beta}(\vec{q}_2,\lambda)\,\epsilon^{\alpha\beta}(\vec{q}_4,\lambda')\big(q_1\cdot q_3 - m^2\big)\cdot 4 \\ + \tfrac{1}{2}\,\epsilon^{\mu\alpha}(\vec{q}_2,\lambda)\,\epsilon^{\nu}{}_{\alpha}(\vec{q}_4,\lambda')\big(q_{1\mu}q_{3\nu} + q_{3\mu}q_{1\nu}\big)\cdot 2\Big\} \\ \times \langle 0|\, \hat{a}_{\vec{p}_2}\hat{a}_{\vec{k}_2\lambda_2}\, \hat{a}^\dagger_{\vec{q}_1}\hat{a}^\dagger_{\vec{q}_2\lambda}\hat{a}_{\vec{q}_3}\hat{a}_{\vec{q}_4\lambda'}\, \hat{a}^\dagger_{\vec{p}_1}\hat{a}^\dagger_{\vec{k}_1\lambda_1}\, |0\rangle\,. \end{aligned} \tag{D.1}$$

The vacuum expectation value fixes $q_1 = p_2$, $q_2 = k_2$, $q_3 = p_1$ and $q_4 = k_1$, and the integration over x gives the delta function in (6.3): the result is (6.7).

D.2 Virtual scalar exchange channels

Inserting the mode expansions into S^{s+u}_{fi}, keeping the terms that annihilate p_1, k_1 and create p_2, k_2, and taking into account all symmetry factors, we find

$$\begin{aligned} S^{s}_{fi} = & -\kappa^2 \int d^4x_1\, d^4x_2 \int \frac{d^4q}{(2\pi)^4} \int \frac{d^3q_1\, d^3q_2\, d^3q_3\, d^3q_4}{(2\pi)^6 \sqrt{16 E_{q_1} E_{q_2} E_{q_3} E_{q_4}}} \sum_{\lambda,\lambda'} e^{i(q_1+q_2)\cdot x_1} e^{-i(q_3+q_4)\cdot x_2} \\ & \times \frac{i\, e^{-iq\cdot(x_1-x_2)}}{q^2 - m^2 + i\varepsilon} \Big[q^\mu q_1^\nu\, \epsilon_{\mu\nu}(\vec{q}_2, \lambda) \Big] \Big[q^\rho q_3^\sigma\, \epsilon_{\rho\sigma}(\vec{q}_4, \lambda') \Big] \\ & \times \langle 0 |\, \hat{a}_{\vec{p}_2} \hat{a}_{\vec{k}_2\lambda_2} \hat{a}^\dagger_{\vec{q}_1} \hat{a}^\dagger_{\vec{q}_2\lambda} \hat{a}_{\vec{q}_3} \hat{a}_{\vec{q}_4\lambda'} \hat{a}^\dagger_{\vec{p}_1} \hat{a}^\dagger_{\vec{k}_1\lambda_1} | 0 \rangle \,, \end{aligned} \tag{D.2}$$

and analogously for S^{u}_{fi} with $k_2 \leftrightarrow -k_1$. Momentum conservation gives

$$q = p_1 + k_1 = p_2 + k_2 \quad (s), \qquad q = p_1 - k_2 = p_2 - k_1 \quad (u), \tag{D.3}$$

therefore

$$q^2 - m^2 = 2p_1\cdot k_1 = 2p_2\cdot k_2 \quad (s), \qquad q^2 - m^2 = -2p_1\cdot k_2 \quad (u). \tag{D.4}$$

We use transversality to simplify the expression:

$$q^\nu \epsilon^{(1)}_{\mu\nu} = (p_1 + k_1)^\nu \epsilon^{(1)}_{\mu\nu} = p_1^\nu \epsilon^{(1)}_{\mu\nu}, \qquad q^\nu \epsilon^{(2)}_{\mu\nu} = (p_2 + k_2)^\nu \epsilon^{(2)}_{\mu\nu} = p_2^\nu \epsilon^{(2)}_{\mu\nu}, \tag{D.5}$$

and we obtain (6.10) and (6.11).

D.3 Graviton self-interaction channel

Term (6) of $\mathcal{V}_3$, Eq. (6.13), does not contribute when the external gravitons are in the TT gauge; only terms (1)–(5) survive. We perform the calculation by contracting each h field with the propagator, one by one. We will repeatedly use, from (3.8),

$$\eta_{\mu\nu}\, G^{\mu\nu,\rho\sigma} = -\eta^{\rho\sigma}, \tag{D.6}$$

and $2p_2^{(\rho} p_1^{\sigma)} \equiv p_2^\rho p_1^\sigma + p_1^\rho p_2^\sigma$.

The last line of each block below is the result after taking the classical (soft-graviton, heavy-scalar) limit.

Term (1).

$$\begin{aligned} \tfrac{1}{2} h^\alpha_\beta \partial^\mu h^\beta_\alpha \partial_\mu h &= \tfrac{1}{2} (\epsilon^{(2)}\cdot\epsilon^{(1)}) \left[q\cdot(k_1 - k_2) \right] \left(\eta_{\alpha\beta} G^{\alpha\beta,\mu\nu} \right) (2 p_{2\mu} p_{1\nu}) \\ &= -(\epsilon^{(2)}\cdot\epsilon^{(1)})\, (p_2\cdot p_1)(-2k_2\cdot k_1) \end{aligned}$$

$$\simeq 2(E\omega)^2(\epsilon^{(2)}\cdot\epsilon^{(1)})(1-\vec{n}_2\cdot\vec{n}_1)$$

Term (2).

$$\begin{aligned}
&-\tfrac{1}{2}h^\alpha_\beta\partial_\alpha h^\mu_\nu\partial^\beta h^\nu_\mu\\
=\;&-\tfrac{1}{2}\Big\{(\epsilon^{(2)}\cdot\epsilon^{(1)})\,(2k_{2\alpha}k_{1\beta}G^{\alpha\beta,\mu\nu})(2p_{2\mu}p_{1\nu})\\
&\quad+2\big[(k_1\cdot\epsilon^{(2)}\cdot q)(\epsilon^{(1)}_{\alpha\beta}G^{\alpha\beta,\mu\nu})(2p_{2\mu}p_{1\nu})-(2\leftrightarrow 1)\big]\Big\}\\
=\;&-\tfrac{1}{2}\Big\{4(\epsilon^{(2)}\cdot\epsilon^{(1)})\Big[\tfrac{1}{2}(p_2\cdot k_2)(p_1\cdot k_1)+\tfrac{1}{2}(p_2\cdot k_1)(p_1\cdot k_2)\\
&\qquad\qquad-\tfrac{1}{2}(k_2\cdot k_1)(p_2\cdot p_1)\Big]\\
&\quad+4\Big[(k_1\cdot\epsilon^{(2)}\cdot q)\big(p_2\cdot\epsilon^{(1)}\cdot p_1\big)-(k_2\cdot\epsilon^{(1)}\cdot q)(p_2\cdot\epsilon^{(2)}\cdot p_1)\Big]\Big\}\\
\simeq\;&-\tfrac{(E\omega)^2}{2}\Big\{4(\epsilon^{(2)}\cdot\epsilon^{(1)})\Big[\tfrac{1}{2}(1-\vec{n}_2\cdot\vec{v})(1-\vec{n}_1\cdot\vec{v})+\tfrac{1}{2}(1-\vec{n}_1\cdot\vec{v})(1-\vec{n}_2\cdot\vec{v})\\
&\qquad\qquad-\tfrac{1}{2}(1-\vec{n}_2\cdot\vec{n}_1)\Big]+O(v^2)\Big\}\\
\simeq\;&-2(E\omega)^2(\epsilon^{(2)}\cdot\epsilon^{(1)})\Big(\tfrac{1}{2}-\vec{n}_2\cdot\vec{v}-\vec{v}\cdot\vec{n}_1+\tfrac{\vec{n}_2\cdot\vec{n}_1}{2}\Big)\\
\simeq\;&-(E\omega)^2(\epsilon^{(2)}\cdot\epsilon^{(1)})\big[1-2\vec{n}_2\cdot\vec{v}-2\vec{v}\cdot\vec{n}_1+\vec{n}_2\cdot\vec{n}_1\big]
\end{aligned}$$

Term (3).

$$\begin{aligned}
&-h^\alpha_\beta\partial_\mu h^\nu_\alpha\partial^\mu h^\beta_\nu\\
=\;&-\Big\{2\,\epsilon^{(2)\,\nu\alpha}\epsilon^{(1)\,\beta}{}_{\nu}\,(k_2\cdot k_1)\,G_{\alpha\beta}{}^{\rho\sigma}(2p_{2\rho}p_{1\sigma})\\
&\quad+\big[\epsilon^{(2)}{}_\beta{}^\alpha\epsilon^{(1)\,\beta\nu}(k_1\cdot q)\,G^{\rho\sigma}{}_{\nu\alpha}(2p_{2\rho}p_{1\sigma})-(2\leftrightarrow 1)\big]\\
&\quad+\big[\epsilon^{(2)}{}_\beta{}^\alpha\epsilon^{(1)}{}_{\alpha\nu}(k_1\cdot q)\,G_{\rho\sigma}{}^{\nu\beta}(2p_2^\rho p_1^\sigma)-(2\leftrightarrow 1)\big]\Big\}\\
=\;&-\Big\{4\Big[\tfrac{1}{2}(p_2\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot p_1)+\tfrac{1}{2}(p_1\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot p_2)\\
&\qquad\qquad-\tfrac{1}{2}(\epsilon^{(2)}\cdot\epsilon^{(1)})(p_2\cdot p_1)\Big](k_2\cdot k_1)\\
&\quad+4\Big[\tfrac{1}{2}(p_2\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot p_1)+\tfrac{1}{2}(p_1\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot p_2)\\
&\qquad\qquad-\tfrac{1}{2}(\epsilon^{(2)}\cdot\epsilon^{(1)})(p_2\cdot p_1)\Big](k_1\cdot q)+\cdots\Big\}\\
\simeq\;&-(E\omega)^2\Big\{-2(\epsilon^{(2)}\cdot\epsilon^{(1)})(1-\vec{n}_2\cdot\vec{n}_1)+4(\epsilon^{(2)}\cdot\epsilon^{(1)})(1-\vec{n}_2\cdot\vec{n}_1)\Big\}\\
\simeq\;&-2(E\omega)^2(\epsilon^{(2)}\cdot\epsilon^{(1)})(1-\vec{n}_2\cdot\vec{n}_1)
\end{aligned}$$

(the omitted terms, denoted $\cdots$, are just repetitions of the terms above).

Term (4).

$$\begin{aligned}\tfrac{1}{4}h\,\partial^\beta h^\mu_\nu \partial_\beta h^\nu_\mu &= \tfrac{1}{4}\Big\{2(k_2\cdot k_1)(\epsilon^{(2)}\cdot\epsilon^{(1)})\big(\eta_{\alpha\beta}G^{\alpha\beta,\mu\nu}\big)(2p_{2\mu}p_{1\nu})\Big\}\\ &\simeq -(E\omega)^2(\epsilon^{(2)}\cdot\epsilon^{(1)})(1-\vec{n}_2\cdot\vec{n}_1)\end{aligned}$$

Term (5).

$$\begin{aligned}&h^\beta_\mu\partial_\nu h^\alpha_\beta\partial^\mu h^\nu_\alpha\\ &= \Big\{(k_1\cdot p_2)(p_1\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot k_2) + (k_1\cdot p_1)(p_2\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot k_2)\\ &\quad - (p_2\cdot p_1)(k_1\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot k_2) + (k_2\cdot p_2)(p_1\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot k_1)\\ &\quad + (k_2\cdot p_1)(p_2\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot k_1) - (p_2\cdot p_1)(k_2\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot k_1)\\ &\quad - (k_1\cdot\epsilon^{(2)}\cdot p_2)(p_1\cdot\epsilon^{(1)}\cdot k_2) - (k_1\cdot\epsilon^{(2)}\cdot p_1)(p_2\cdot\epsilon^{(1)}\cdot k_2)\\ &\quad + (p_2\cdot p_1)(k_1\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot k_2) - (k_2\cdot\epsilon^{(1)}\cdot p_2)(p_1\cdot\epsilon^{(2)}\cdot k_1)\\ &\quad - (k_2\cdot\epsilon^{(1)}\cdot p_1)(p_2\cdot\epsilon^{(2)}\cdot k_1) + (p_2\cdot p_1)(k_2\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot k_1)\\ &\quad + (k_1\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot p_2)(p_1\cdot k_1) + (k_1\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot p_1)(p_2\cdot k_1)\\ &\quad + (k_2\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot p_2)(p_1\cdot k_2) + (k_2\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot p_1)(p_2\cdot k_2)\Big\}\\ &\simeq -(E\omega)^2\Big\{2\,v\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot n_2 + 2\,v\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot n_1 + 2\,n_1\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot v + 2\,n_2\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot v\Big\}\\ &\simeq -(E\omega)^2\Big\{4\,v\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot n_2 + 4\,v\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot n_1\Big\}\end{aligned}$$

(the $(p_2\cdot p_1)$ terms cancel exactly, and the products in the third and fourth lines are $O(v^2)$).

Summing everything, the $\vec{n}_2\cdot\vec{n}_1$ pieces cancel exactly:

$$\begin{aligned}\mathcal{V}_3 \simeq -2(E\omega)^2\Big\{&(\epsilon^{(2)}\cdot\epsilon^{(1)})\big[1-\vec{v}\cdot(\vec{n}_2+\vec{n}_1)\big]\\ &+2\big(\vec{v}\cdot\epsilon^{(2)}\cdot\epsilon^{(1)}\cdot\vec{n}_2+\vec{v}\cdot\epsilon^{(1)}\cdot\epsilon^{(2)}\cdot\vec{n}_1\big)\Big\},\end{aligned}\tag{D.7}$$

from which (6.15) follows, with $\mathcal{M}$ the term in curly brackets.

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E DISCARDED Open Quatum Systems

We will develop the theory of quantum open systems from the point of view of the so- called Feynman–Vernon influence functional. This approach and its closely related Schwinger–Keldysh or closed time path method will underlie the anal- ysis of nonequilibrium quantum fields.

We start from the path integral representation of the unitary evolution operator $U(t)$:

$$\langle x, t \mid U(t) \mid x_0, 0\rangle = \int_{x(0)=x_0}^{x(t)=x} \mathcal{D}x \; e^{\frac{i}{\hbar}S[x]}$$

Since in open quantum systems we are interested in density matrix ρ:

$$\rho(x, x', t) = \langle x \mid U(t)\, \rho\, U^{\dagger}(t) \mid x'\rangle \tag{E.1}$$

$$= \int_{x(0)=x_i}^{x(t)=x} \mathcal{D}x \int_{x'(0)=x_i'}^{x'(t)=x'} \mathcal{D}x' \; \exp\left\{\frac{i}{\hbar}\big(S[x] - S[x']\big)\right\} \; \rho\big(x_i, x_i', 0\big). \tag{E.2}$$

Since the hermitian coniugate of the evolution operators evolves the bra and not the ket. What we have is the evolution of two histories rather than one. Equivalently, we could think of a unique path integral, with a forward evolution then a backward one. The reason we speak of backward evolution is the sign of the action, which leds to anti-time ordering of the operators. In path integral QFT, we recover propagators by integrating over paths two fields. If we do the same here what we get is really interesting: defining what field we are integrating, we get different evolutions.

$$\begin{aligned} G_{11}(\tau, \tau') &= \int \mathcal{D}x\, \mathcal{D}x'\, e^{i(S[x]-S[x'])}\, \rho(x(0), x'(0), 0)\; x(\tau)\, x(\tau') \\ &\equiv \left\langle \hat{X}(\tau)\hat{X}(\tau')\right\rangle . \end{aligned} \tag{E.3}$$

The reason is that we can decompose the path integral over times:

$$G_{11}(\tau, \tau') = \int dx(0)\, dx'(0)\, dx(\tau')\, dx(\tau)\, dx(t)$$

$$\times \int_{x(0)}^{x(\tau')} \mathcal{D}x\, e^{iS[x]}\, x(\tau') \int_{x(\tau')}^{x(\tau)} \mathcal{D}x\, e^{iS[x]}\, x(\tau)$$
$$\times \int_{x(\tau)}^{x(t)} \mathcal{D}x\, e^{iS[x]} \int_{x'(0)}^{x(t)} \mathcal{D}x'\, e^{-iS[x']}\, \rho(x(0), x'(0), 0)\,. \tag{E.4}$$

And from the traces we get the starting results:

$$G_{11}(\tau,\tau') = \int dx(0)\, dx'(0)\, dx(\tau')\, dx(\tau)\, dx(t)$$
$$\times \langle x(t)|U(t,\tau)|x(\tau)\rangle\, x(\tau)\, \langle x(\tau)|U(\tau,\tau')|x(\tau')\rangle\, x(\tau')$$
$$\times \langle x(\tau')|U(\tau',0)|x(0)\rangle\, \langle x(0)|\rho(0)|x'(0)\rangle\, \langle x'(0)|U(0,t)|x(t)\rangle. \tag{E.5}$$

$$G_{11}(\tau,\tau') = \mathrm{Tr}\big[U(t,\tau)\,\hat{X}\,U(\tau,\tau')\,\hat{X}\,U(\tau',0)\,\rho(0)\,U(0,t)\big] \equiv \big\langle \hat{X}(\tau)\hat{X}(\tau')\big\rangle\,. \tag{E.6}$$

For G^{22} we get the same result but with time ordering reversed, since we could make the substitution $t \to -t$ to recover a positive action. On the other hand:

$$G_{12}(\tau,\tau') = \int \mathcal{D}x\, \mathcal{D}x'\; e^{i(S[x]-S[x'])}\, \rho\big(x(0), x'(0), 0\big)\, x(\tau)\, x'(\tau') \quad \text{con } x(t) = x'(t). \tag{E.7}$$

$$G_{12}(\tau,\tau') = \mathrm{Tr}\Big[U(0,\tau')\,\hat{X}\,U(\tau',t)\,U(t,\tau)\,\hat{X}\,U(\tau,0)\,\rho(0)\Big]. \tag{E.8}$$

Primed operators, so primed fields, come always on the left.

F DISCARDED Influence Functional

F.1 Reduced density matrix and definition of the influence functional

$$\rho(x_q,q,t) = \langle x_q,q,t|\rho|x'_q,q,t\rangle$$
$$= \int dx_i\, dq_i\, dx'_i\, dq'_i\; \langle x_q,q,t|x_i,q_i,0\rangle\, \langle x_i,q_i,0|\rho|x'_i,q'_i,0\rangle\, \langle x'_i,q'_i,0|x'_q,q,t\rangle$$
$$= \int dx_i\, dq_i\, dx'_i\, dq'_i\; \left(\int_{x_i,q_i}^{x_q,q} \mathcal{D}x\, \mathcal{D}q\; e^{iS[x,q]}\right) \rho(x_i,q_i,x'_i,q'_i,0) \left(\int_{x'_i,q'_i}^{x'_q,q} \mathcal{D}x'\, \mathcal{D}q'\; e^{-iS[x',q']}\right)$$
$$\equiv \int dx_i\, dq_i\, dx'_i\, dq'_i\; \mathcal{J}\,(xq, x'q',\, t\,|\,x_iq_i, x'_iq'_i, 0)\, \rho(x_i,q_i,x'_i,q'_i,0)$$

$$\rho_r(x,x',t) = \int_{-\infty}^{+\infty} dq\; \rho(x,q,x',q,t).$$

$$\rho(x_i, q_i, x_i', q_i', 0) = \rho_S(x_i, x_i', 0)\,\rho_E(q_i, q_i', 0) \quad \Rightarrow \quad \rho_r(x, x', t) = \int dx_i\, dx_i'\, J_r(x, x', t|x_i, x_i', 0)\, \rho_S(x_i, x_i', 0$$

$$J_r(x, x', t|x_i, x_i', 0) = \int_{x_i}^{x} \mathcal{D}x \int_{x_i'}^{x'} \mathcal{D}x'\, \exp\big(iS[x] - iS[x']\big)\, F[x, x'].$$

$$F[x, x'] = \exp\big(iS_{\rm IF}[x, x']\big), \qquad F[x, x'] = \int dq_i\, dq_i'\, dq\, \rho_E(q_i, q_i', 0) \int_{q_i}^{q} \mathcal{D}q\, e^{i(S_E[q]+S_{\rm int}[x,q])} \int_{q_i'}^{q} \mathcal{D}q'$$

F.2 General properties of the influence action

$$e^{iS_{\rm IF}[x,x',t]} = \mathrm{Tr}\left\{U_{x'}(0,t)\, U_x(t,0)\, \rho_E(0)\right\}.$$

$$S_{\rm IF}[x, x, t] = 0, \qquad S_{\rm IF}[x', x, t] = -S_{\rm IF}^*[x, x', t].$$

$$u(t) = x(t) - x'(t), \qquad X(t) = \frac{x(t) + x'(t)}{2},$$

$$S_{\rm IF}[X, u, t] = \sum_{k=1}^{\infty} \frac{1}{k!} \int dt_1 \cdots dt_k\, S^{(k)}[X; t_1, \ldots, t_k; t]\, u(t_1) \cdots u(t_k),$$

odd orders real, even orders imaginary.

$$\left.\frac{\delta S_{\rm IF}}{\delta x}\right|_{x=x'} = -\left.\frac{\delta S_{\rm IF}}{\delta x'}\right|_{x=x'}.$$

CTP doublet:

$$x^a(t), \quad a = 1, 2, \quad x^1 = x,\ x^2 = x', \qquad c_{ab} = \mathrm{diag}(1, -1), \quad c^{ab} = \mathrm{diag}(1, -1),$$

$$\int dt\, c_{ab}\, \dot{x}^a \dot{x}^b = \int dt \left(\dot{x}^2 - \dot{x}'^2\right), \qquad S[x^a] = S[x] - S[x'].$$

F.3 Application to linearized gravity: the graviton as environment

System: ϕ. Environment: h_{ij}.

F.3.1 Graviton action

$$S_{HE}^{(2)} = -\frac{1}{2} \int d^4x\, d^4x'\, h^{ij}(x)\, L_{ij}^{kl}(x, x')\, h_{kl}(x'), \qquad L_{ij}^{kl} = \frac{1}{\kappa^2}\, \delta^{(4)}(x - x')\, \eta_i^k \eta_j^l\, \Box.$$

$$S_I^{(1)} = \int d^4x\, h^{ij} B_{ij}, \qquad B_{ij} = -\tfrac{1}{2} \partial_i \phi\, \partial_j \phi.$$

There is also $S_I^{(2)} = -\frac{1}{2}\int h\,C\,h$, but at leading order $M = L + C \approx L$: C is dropped from here on.

$$F[x,x'] = e^{iS_{\rm IF}[\phi,\phi']} = \int dh\; dh_i\; dh_i'\; \rho_E(h_i,h_i',0)\; e^{-iA[T\langle hh\rangle,\phi]}\; e^{-iA'[T\langle h'h'\rangle,\phi']} \sim e^{-\frac{1}{2}\int d^4x\; h^{ij}C_{ij}^{kl}h_{kl}}.$$

Here h_i, h_i' still denote the two copies of the environmental path integral prior to integrating out the initial state — temporary use of the prime, distinct from the convention adopted below. Switch to the CTP formalism (Calzetta): the influence action is the functional Fourier transform of a Gaussian functional of the bath histories, hence remains Gaussian.

F.3.2 General Gaussian result for a linear coupling

Convention: from here on h_{ij} and Q are never primed — they are the same operator in the Heisenberg picture, and the CTP branch is encoded in the ordering of the product (T, $\bar{T}$, Wightman), not in a second field. The prime (or ±, cl/Δ) is reserved for system variables: $x, x', \phi, \phi', B, B'$.

$$S_{\rm IF}[x_a] = \frac{1}{2}\int dt\,dt'\,x_a(t)\,M_{ab}(t,t')\,x_b(t').$$

$$M_{ab}(t,t') = -i\,\frac{\delta^2}{\delta x_a(t)\,\delta x_b(t')}e^{iS_{\rm IF}[x_a]}\bigg|_{x_a=0},$$

$$\frac{\delta^2}{\delta x_a(t)\,\delta x_b(t')}e^{iS_{\rm IF}[x_a]}\bigg|_{x_a=0} = -\frac{1}{2}\int \mathcal{D}q\; e^{iS_E[q]}\,Q_a(t)\,Q_b(t')\,\rho_E(q_1(0),q_2(0),0), \qquad q_1(T)=q_2(T).$$

$$M_{ab}(t,t') = \begin{pmatrix} \langle T\,[Q(t)Q(t')]\rangle & -\langle[Q(t')Q(t)]\rangle \\ -\langle[Q(t)Q(t')]\rangle & \langle\tilde{T}\,[Q(t)Q(t')]\rangle \end{pmatrix}.$$

$$\frac{i}{2}S_{\rm IF} = \frac{i}{2}\int dt\,dt'\,\Big[\langle TQ(t)Q(t')\rangle\,x(t)x(t') - \langle Q(t')Q(t)\rangle\,x(t)x'(t') - \langle Q(t)Q(t')\rangle\,x'(t)x(t') + \langle\tilde{T}Q(t)Q(t')\rangle\,x$$

With $x = X + u/2$, $x' = X - u/2$:

$$S_{\rm IF} = \int dt\,dt'\,u(t)\,D(t,t')\,X(t') + \frac{i}{2}\int dt\,dt'\,u(t)\,N(t,t')\,u(t'),$$

$$D(t,t') = i\,\theta(t-t')\,\langle[Q(t),Q(t')]\rangle, \qquad N(t,t') = \frac{1}{2}\,\langle\{Q(t),Q(t')\}\rangle.$$

F.3.3 Graviton propagator

$$(L^{-1})_{ij}^{kl} = -i\kappa^2\,\delta^{(4)}(x-x')\,\eta_i^k\eta_j^l\,\Box^{-1} \quad\Longrightarrow\quad \langle 0|T\{h_{ij}(x)h^{kl}(x')\}|0\rangle = i(L^{-1})_{ij}^{kl} = \kappa^2\,\eta_i^k\eta_j^l\,\Box^{-1}\delta^{(4)}(x-x').$$

F.3.4 Applying the general result to h and $B_{ij}(\phi)$

$$M_{ab}^{ijkl}(t,t') = i\Big[\langle T[h^{ij}(t)h^{kl}(t')]\rangle - \langle h^{kl}(t')h^{ij}(t)\rangle - \langle h^{ij}(t)h^{kl}(t')\rangle + \langle \bar{T}[h^{ij}(t)h^{kl}(t')]\rangle\Big] \tag{F.1}$$

$$S_{\rm IF} = \frac{i}{2}\int dt\,dt'\Big[\langle T[h^{ij}(t)h^{kl}(t')]\rangle\, B_{ij}(t)B_{kl}(t') - \langle h^{kl}(t')h^{ij}(t)\rangle\, B_{ij}(t)B'_{kl}(t') \tag{F.2}$$

$$- \langle h^{ij}(t)h^{kl}(t')\rangle\, B'_{ij}(t)B_{kl}(t') + \langle \bar{T}[h^{ij}(t)h^{kl}(t')]\rangle\, B'_{ij}(t)B'_{kl}(t')\Big]. \tag{F.3}$$

Note: h is never primed; the system branch is carried only by B, B'.

F.3.5 Classical and difference variables

With $\phi_\pm = \phi_{\rm cl} \pm \phi_\Delta/2$, $B^\pm_{ij} = -\frac{1}{2}\partial_i\phi_\pm\partial_j\phi_\pm$:

$$X_B^{ij} = -\frac{1}{2}\partial^i\phi_{\rm cl}\partial^j\phi_{\rm cl} - \frac{1}{8}\partial^i\phi_\Delta\partial^j\phi_\Delta, \qquad u_B^{ij} = -\Big(\partial^i\phi_{\rm cl}\partial^j\phi_\Delta + \partial^i\phi_\Delta\partial^j\phi_{\rm cl}\Big).$$

$$S_{\rm IF} = \int dt\,dt'\; u_B^{ij}(t)\, D_{ijkl}(t,t')\, X_B^{kl}(t') + \frac{i}{2}\int dt\,dt'\; u_B^{ij}(t)\, N_{ijkl}(t,t')\, u_B^{kl}(t'),$$

$$D_{ijkl}(t,t') = i\,\theta(t-t')\,\langle[h_{ij}(t), h_{kl}(t')]\rangle, \qquad N_{ijkl}(t,t') = \tfrac{1}{2}\langle\{h_{ij}(t), h_{kl}(t')\}\rangle.$$

G DISCARDED Open quantum systems: CTP path integral

Basis-independent form:

$$F[x,x'] = \mathrm{Tr}\big[U(x;t)\,\rho_E(0)\,U^\dagger(x';t)\big] = \int dq\,dq_i\,dq_i'\,\rho_E(q_i,q_i';0)\,\langle q|U(x;t)|q_i\rangle\,\langle q|U^\dagger(x';t)|q_i'\rangle. \tag{G.1}$$

Forward and conjugate amplitudes:

$$\langle q|U(x;t)|q_i\rangle = \int_{q_i}^{q}\mathcal{D}q_+\,e^{i(S_E[q_+]+S_{\rm int}[x,q_+])}, \qquad \langle q|U^\dagger(x';t)|q_i'\rangle = \int_{q_i'}^{q}\mathcal{D}q_-\,e^{-i(S_E[q_-]+S_{\rm int}[x',q_-])}.$$

The minus sign in the exponent is not a different dynamics: it comes from complex conjugation of the amplitude. The two integrations combine into a single path integral over the contour $C = C_+ \cup C_-$ (forward then backward in time, glued at t_f), with free endpoints q_i, q_i' weighted by ρ_E:

$$F[x,x'] = \int dq_i\,dq_i'\,\rho_E(q_i,q_i';0)\int_C \mathcal{D}q\;e^{iS_C[q;x,x']}, \qquad S_C = S_E[q_+]+S_{\rm int}[x,q_+]-\big(S_E[q_-]+S_{\rm int}[x',q_-]\big) \tag{G.2}$$

The only physical difference between the two branches is the operator time ordering.

H DISCARDED Open quantum systems: Gaussian environment

Origin of the matrix structure

The environment field is doubled, $q \to (q_+, q_-)$, $J \to (J_+, J_-)$, contour metric $C = \mathrm{diag}(1,-1)$: a direct consequence of $S[q_+] - S[q_-]$, not an extra assumption.

Gaussian integral, one-component case

$$Z[J] = \int dq\;\exp\left[\frac{i}{2}qKq + iJq\right] = \mathcal{N}\exp\left[-\frac{i}{2}JK^{-1}J\right]. \tag{H.1}$$

CTP case

$$S[q_+] - S[q_-] = \tfrac{1}{2}\,q_aK_{ab}q_b, \qquad K_{ab}(x,x') = \delta^{(4)}(x-x')\,\mathrm{diag}(K,-K). \tag{H.2}$$

K_{ab} is diagonal, but $G_{ab} = (K_{ab})^{-1}$ is not: inverting it is a differential problem that requires boundary conditions (continuity at $t_{\max}$, correlations at t_i fixed by ρ), which generate the

off-diagonal, state-dependent components:

$$G_{ab} = \begin{pmatrix} \langle T[q(x)q(x')]\rangle & \langle q(x')q(x)\rangle \\ \langle q(x)q(x')\rangle & \langle \bar{T}[q(x)q(x')]\rangle \end{pmatrix} = \begin{pmatrix} G_F & G^< \\ G^> & G_{\bar{F}} \end{pmatrix}. \tag{H.3}$$

Influence functional for a Gaussian environment

Quadratic environment action, Gaussian ρ_e, bilinear coupling $S_{\rm int} = \int dt\, x_a(t)\, Q_a[q(t)]$ with Q_a linear in q. The functional Fourier transform of a Gaussian is Gaussian, so S_{IF} is quadratic in x_+, x_-:

$$S_{IF} = \frac{1}{2}\int dt\, dt'\; x_a(t)\, M_{ab}(t,t')\, x_b(t'), \qquad M_{ab}(t,t') = -i\,\frac{\delta^2}{\delta x_a(t)\,\delta x_b(t')}\, e^{iS_{IF}[x_a]}\Big|_{x_a=0}. \tag{H.4}$$

Direct variation gives a CTP expectation value of Q with the system-environment coupling switched off, i.e. exactly G_{ab} above, evaluated on Q and traced with ρ_e:

$$M_{ab}(t,t') = i\,G^{Q}_{ab}(t,t') = i\begin{pmatrix} \langle T[Q(t)Q(t')]\rangle & -\langle Q(t')Q(t)\rangle \\ -\langle Q(t)Q(t')\rangle & -\langle \bar{T}[Q(t)Q(t')]\rangle \end{pmatrix}. \tag{H.5}$$

Explicitly, with $x_\pm \equiv x_\pm(t)$, $x'_\pm \equiv x_\pm(t')$:

$$S_{IF} = \frac{i}{2}\int dt\, dt' \Big[\langle T[Q(t)Q(t')]\rangle\, x_+x'_+ - \langle Q(t')Q(t)\rangle\, x_+x'_- - \langle Q(t)Q(t')\rangle\, x_-x'_+ + \langle \bar{T}[Q(t)Q(t')]\rangle\, x_-x'_-\Big]. \tag{H.6}$$

The averages are taken with $Q(t)$ evolving under the free environment Hamiltonian, decoupled from the system — together with Gaussianity of ρ_e, this is what keeps M_{ab} closed-form.

Rotation to dissipation and noise

Keldysh basis $X = (x_+ + x_-)/2$, $u = x_+ - x_-$:

$$S_{IF} = \int dt\, dt' \,\Big[u(t)\, D(t,t')\, X(t') + \frac{i}{2}\, u(t)\, N(t,t')\, u(t')\Big], \tag{H.7}$$

$$D(t,t') = i\,\theta(t-t')\,\big\langle [Q(t), Q(t')]\big\rangle, \qquad N(t,t') = \frac{1}{2}\,\big\langle \{Q(t), Q(t')\}\big\rangle. \tag{H.8}$$

I DISCARDED Full linearized theory

Expanding the Einstein–Hilbert action around a flat background generates an infinite series of nonlinear self–interaction terms for the gravitational field. At linear order, the equations of motion take the form

$$\mathcal{E}^{\alpha\beta}_{\mu\nu} h_{\alpha\beta} = \frac{\kappa}{2} T_{\mu\nu},$$

where $\mathcal{E}^{\alpha\beta}_{\mu\nu}$ denotes the linearized Einstein operator.

With the allowed gauges, matter fields satisfy

$$\partial^{\mu} T_{\mu\nu} = 0,$$

However, expanding higher orders in κ, the gravitational field contributes to its own source. The equations of motion can be written schematically as

$$\mathcal{E}^{\alpha\beta}_{\mu\nu} h_{\alpha\beta} = \frac{\kappa}{2} \left(T_{\mu\nu} + t_{\mu\nu}[h] \right), \tag{I.1}$$

where $t_{\mu\nu}[h]$ is the energy–momentum pseudotensor arising from the nonlinearities of the Einstein–Hilbert action. It is object constructed order by order.

In particular, the contribution at order i in the expansion is sourced by the term $S^{(i+1)}[h]$ in the action. For example,

$$t^{(2)}_{\mu\nu} \sim \frac{\delta S^{(3)}[h]}{\delta h^{\mu\nu}}. \tag{I.2}$$

This structure shows that the graviton couples to its own effective stress–energy, leading to a nonlinear "bootstrap" mechanism: iterating the cubic and higher–order vertices reconstructs the full Einstein theory.

Gauge invariance at linear order is expressed by the infinitesimal diffeomorphism

$$\delta^{(0)}_{\xi} h_{\mu\nu} = \partial_{\mu}\xi_{\nu} + \partial_{\nu}\xi_{\mu}, \tag{I.3}$$

while at nonlinear order the full transformation must include terms like $\sim h\partial\xi$, which is the complete expansion of the diffeomorfism.

In summary, the self–interaction structure of the graviton encodes the nonlinear completion of the theory, and general covariance emerges as the perturbative realization of full diffeomorphism invariance order by order in κ.

J Discarded Calculus: TT Projector self int

The contribute from this interaction will be in the form:

$$\mathcal{M}_{hhh} = \left(k_1^{\mu} k_3^{\nu} + k_1^{\nu} k_3^{\mu} - \eta^{\mu\nu}(k_1 \cdot k_3 - m^2) \right) \frac{\Lambda_{\mu\nu,\alpha\beta}(q)}{q^2 + i\epsilon} V^{\alpha\beta}(k_2, \epsilon^{(2)}; k_4, \epsilon^{(4)})$$

It's easier to project the left term with the TT projector, which actually has only spatial

components ($\alpha\beta = 1, 2, 3$).

$$\left(k_1^\mu k_3^\nu + k_1^\nu k_3^\mu - \eta^{\mu\nu}(k_1 \cdot k_3 - m^2)\right)\frac{\Lambda_{\mu\nu}^{\ \ \alpha\beta}}{q^2 + i\epsilon} = \tag{J.1}$$

$$\frac{1}{q^2 + i\epsilon}\Big[k_1^\alpha k_3^\beta + k_1^\beta k_3^\alpha - (\vec{k_1} \cdot \vec{k_3})(\hat{q}_\alpha \hat{q}_\beta - \delta_{\alpha\beta}) \tag{J.2}$$

$$- (\vec{k_3} \cdot \hat{q})(k_1^\alpha q^\beta + k_1^\beta q^\alpha)) - (\vec{k_1} \cdot \hat{q})(k_3^\alpha q^\beta + k_3^\beta q^\alpha) \tag{J.3}$$

$$+ (\vec{k_1} \cdot \hat{q})(\vec{k_3} \cdot \hat{q})\ \left(\delta^{\alpha\beta} + q^\alpha q^\beta\right)\ \Big] \tag{J.4}$$

At this point, we only have to consider all of Wick contractions with $V^{\alpha\beta}$, the Vortex is:

$$\text{Vortex} \ = \ h_\rho^{\ \sigma}(x_2)\left[\frac{1}{2}\partial_\sigma h_\nu^{\ \mu}(x_2)\partial^\rho h_\mu^{\ \nu}(x_2) + \partial_\mu h_\sigma^{\ \nu}(x_2)\partial^\mu h_\nu^{\ \rho}(x_2) - \partial_\nu h_\sigma^{\ \alpha}(x_2)\partial^\rho h_\alpha^{\ \nu}(x_2)\right] \tag{J.5}$$

For simplicity, we will decompose into 3 contributes depending on wheter we are contracting the first, second or third term: $V = \mathcal{A}^{(1)} + \mathcal{A}^{(2)} + \mathcal{A}^{(3)}$. What happens is we free the vertex of a term, which will form the propagator, and express everything in terms of the indexed $\mu\nu$. Then, we will pose the on-shell conditions on the free legs:

$$\begin{aligned}\mathcal{A}^{(1)\alpha\beta} = &-(k_2)^\beta (k_4)^\alpha\ (\epsilon_2 \cdot \epsilon_4) - 2(k_2 \cdot k_4)\ (\epsilon_2 \cdot \epsilon_4)^{\beta\alpha} \\ &+ (k_4)^\alpha\ (\epsilon_2 \cdot \epsilon_4 \cdot k_2)^\beta + (k_2)^\alpha\ (\epsilon_4 \cdot \epsilon_2 \cdot k_4)^\beta\end{aligned} \tag{J.6}$$

$$\begin{aligned}\mathcal{A}^{(2)\alpha\beta} = &-\frac{1}{2}(k_4 \cdot \epsilon_2 \cdot q)\ \epsilon_4^{\alpha\beta} - \frac{1}{2}(k_2 \cdot \epsilon_4 \cdot q)\ \epsilon_2^{\alpha\beta} \\ &- (\epsilon_2 \cdot \epsilon_4)^{\alpha\beta}\, q \cdot (k_2 + k_4) + (\epsilon_2 \cdot k_4)^\alpha (\epsilon_4 \cdot q)^\beta + (\epsilon_4 \cdot k_2)^\alpha (\epsilon_2 \cdot q)^\beta,\end{aligned} \tag{J.7}$$

$$\begin{aligned}\mathcal{A}^{(3)\alpha\beta} = &-\frac{1}{2}(q \cdot \epsilon_2 \cdot k_4)\ \epsilon_4^{\alpha\beta} - \frac{1}{2}(q \cdot \epsilon_4 \cdot k_2)\ \epsilon_2^{\alpha\beta} \\ &- (\epsilon_2 \cdot \epsilon_4)^{\alpha\beta}(q \cdot k_4) - (\epsilon_4 \cdot \epsilon_2)^{\alpha\beta}(q \cdot k_2) \\ &+ (\epsilon_4 \cdot \epsilon_2 \cdot q)^\alpha k_4^\beta + (\epsilon_2 \cdot \epsilon_4 \cdot q)^\alpha k_2^\beta.\end{aligned} \tag{J.8}$$

Summing all up:

$$\begin{aligned}\mathcal{A}^{(1)\alpha\beta} = &-(k_2)^\beta (k_4)^\alpha\ (\epsilon_2 \cdot \epsilon_4) - 2(k_2 \cdot k_4)\ (\epsilon_2 \cdot \epsilon_4)^{\beta\alpha} \\ &+ (k_4)^\alpha\ (\epsilon_2 \cdot \epsilon_4 \cdot k_2)^\beta + (k_2)^\alpha\ (\epsilon_4 \cdot \epsilon_2 \cdot k_4)^\beta\end{aligned} \tag{J.9}$$

$$\mathcal{A}^{(2)\alpha\beta} = -(k_4 \cdot \epsilon_2 \cdot q)\ \epsilon_4^{\alpha\beta} - (k_2 \cdot \epsilon_4 \cdot q)\ \epsilon_2^{\alpha\beta} \tag{J.10}$$

$$\mathcal{A}^{(3)\alpha\beta} = +(\epsilon_4\cdot\epsilon_2\cdot q)^\alpha k_4^\beta + (\epsilon_2\cdot\epsilon_4\cdot q)^\alpha k_2^\beta. \tag{J.11}$$